Building AI Evaluation Benchmarks for Australian Contract Law

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Part 1: General Literature Review Abstract

1 Introduction

The Australian legal environment is facing significant challenges due to the increasing workload and case loading in courts. Take Victoria's supreme court pending as an example, the pending courts were constantly increasing after 2021 (Figure 1), which can impact the timely delivery and the quality of justice. The disputes and the administrative tasks of law are growing in complexity and volume, courts become overwhelmed, and end in delays in case processing and resolution. Taking the USA as an example, the high volume of cases throughout the court system has led to substantial backlogs, raising concerns about access to timely justice (Zheng et al., 2024). Similarly, the legal system in Europe faces similar challenges, this issue has been made worse by the complexity of the process and the sheer volume of legal documentation (Zheng et al., 2024). In Australia, the circumstances share the same issue as limited resources and growing case volumes strain the judicial system, highlighting the need for innovative legal solutions (Court Service Victoria, 2024).

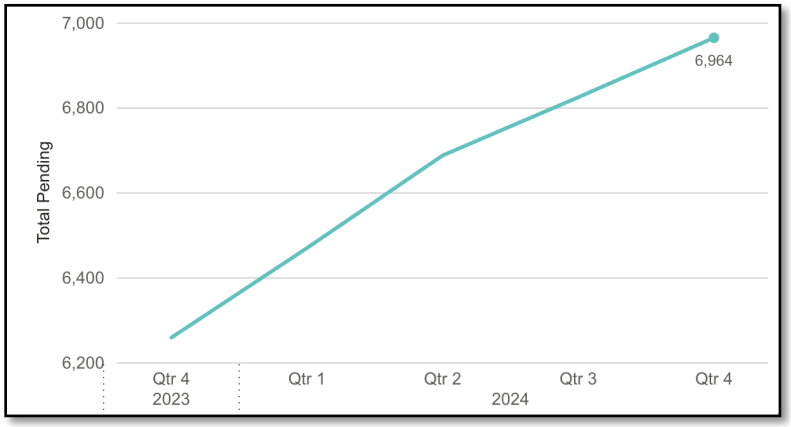

Figure 1: Victoria Supreme Court pending

Another obvious issue in the legal domain is the high cost of legal consultation or service (Olling, 2025), which acts as an obstacle to justice, especially for vulnerable and low-income groups. A considerable number of individuals and small businesses lack the financial resources to procure competent legal counsel. This predicament has led to an escalating prevalence of self-representation, which frequently results in unfavourable outcomes.

In light of these challenges, the growth of Generative AI (GenAI) technologies, especially Large Language Models (LLMs), has shown the potential to revolutionize the legal sector (Kang et al., 2023). One of the advantages of implementing LLMs into legal practice is the potential for cost reduction with the efficiency increase. By automating routine tasks, legal professionals can minimize the time spent on document review, contract analysis (Hendrycks et al., 2021), cases categorization, and most importantly lowering service costs (Palos, 2025; Holzenberger et al., 2020). Efforts to adapt language models for legal applications date back to at least 2020, when domain-specific models

such as LEGAL-BERT were introduced to address the unique linguistic and contextual challenges of legal text (Chalkidis et al., 2020; Holzenberger & Van Durme, 2021).
To methodically evaluate the efficacy and the performance of LLMs in legal settings, a variety of benchmarks have been developed in recent years. LegalBench (Guha et al., 2023) is a prominent instance of such a benchmark, evaluating models on tasks such as legal reasoning, rule application, rhetorical analysis, legal information extraction, and so. Similarly, LexGLUE (Chalkidis et al., 2022) provides a structured evaluation of LLMs, focusing on tasks such as legal judgment prediction, case summarization, and so on. However, it should be noted that the majority of these benchmarks are primarily focused on US and EU law, thereby creating a notable gap in evaluating LLMs against Australian legal contexts. This gap is particularly significant because Australia's legal system and jurisprudence differ significantly from those of the United States and the European Union.

To date, the Allens AI Australian Law Benchmark (Allen, 2024) marks an early exploration into LLM evaluation within Australian law. It provided valuable early insights into the strengths and limitations of prominent models such as GPT-4. However, the scope of the benchmark is relatively limited, covering only 30 questions across a few core legal domains, such as contract law, tax, privacy, and other areas of legal practice – which means that it does not yet provide a sufficiently comprehensive reference for legal practitioners or researchers. In particular, there is a key opportunity to construct benchmarks that are deeply aligned with the Australian jurisdictional nuances and broader informational demands of the legal profession.
In summary, the objective of this thesis is to identify and implement a benchmark tailored to the Australian legal system, commencing with the domain of contract law. First, we aim to provide an overview of existing legal benchmarks developed outside the Australian legal context, analysing their structures, tasks, and evaluation methods. Second, we seek to identify appropriate data sources and areas of law for developing an Australian law-specific benchmark, with a particular focus on the AustLII corpus and the domain of contract law. Third, we propose to test and evaluate the performance of state-of-the-art LLMs against this Australian-specific benchmark to assess their reliability in addressing legal tasks within the Australian jurisdiction. This study addresses a gap in current legal AI research by laying the groundwork for a realistic, domain-focused evaluation framework, contributing to the development of responsible and context-aware legal AI systems for future research and applications.

2 Substantive Literature Review

2.1 Legal AI Benchmark

The rapid evolution of LLMs has opened the way for their integration into various legal applications, including legal research, document analysis, and even preliminary legal advice. To ensure the reliability and efficacy of these models, several benchmarking efforts have been initiated, focusing on evaluating their performance on specific legal tasks. Among the most well-known benchmarking initiatives are those that comprehensively assess the capabilities of LLMs in legal contexts, including LegalBench, LegalAgentBench, and benchmarks focused on specific regions such as Australia.

2.1.1 LegalBench (Guha et al., 2023)

One of the most comprehensive and widely referenced benchmarks is LegalBench, this framework provides a standardized approach to evaluating LLMs in legal natural language processing (NLP) tasks. LegalBench covers several tasks, including judgment prediction, legal information extraction, case summarization, and rule application. This benchmark facilitates a comparative analysis of diverse models, including GPT-4, LLaMa, and Claude, by assessing their capacity to execute critical legal functions. The benchmark's design ensures that models are evaluated not only for their accuracy but also for their contextual understanding and consistency when dealing with legal texts. By offering a structured evaluation, LegalBench has become a crucial tool in assessing the robustness of LLMs in legal domains.

The development of LegalBench involved a collaborative effort between legal professionals and AI researchers, ensuring that it addresses practical legal reasoning capabilities. In contrast to prior benchmarks, it emphasizes not only accuracy but also contextual relevance, rendering it more suitable for the intricacies inherent in legal tasks. In practice, this means assessing whether LLMs can not only provide correct answers, but also justify those answers based on existing legal principles. The incorporation of multi-task assessments serves to further strengthen its relevance, as legal reasoning often requires synthesizing information from multiple sources. LegalBench has garnered recognition for its proficiency in evaluating a wide range of legal tasks, such as information extraction and reasoning analysis. To support this analysis, a detailed summary of benchmark tasks from each paper has been compiled (Table 1). For complete task descriptions from each paper, please refer to Appendix A to F.

LEGALBENCH					
Task	**Variations?**	**Description of the task**	**Evaluation Metrics**	**Jurisdictions (Countries)**	**Task format (MCQ, T/F, Short/Long answers)**
cuad_joint_ip_ownership	Yes	Does the clause provide for joint or shared ownership of intellectual property between the parties to the contract?	Binary classification (Accuracy, F1)	US / General	Binary classification
oral_argument_question_purpose	Yes	evaluatesanLLM'sabilitytoidentifythepurposethataparticularquestion(from Supreme Court oral arguments) plays	Classification accuracy	US	MCQ
citation_prediction_open	Yes	Predict appropriate legal citation's for a legal statement.	Citation accuracy, F1	US	Short answer (Open)
learned_hands_benefits	Yes	Determine whether a user-submitted post raises issues about public benefits or social services (e.g., food stamps, housing, etc.).	Multi-label accuracy	US	Binary classification (Yes/No)

Table 1: Summary of LegalBench Tasks Across Referenced Datasets

2.1.2 LegalBench-RAG (Zheng et al., 2024)

The LegalBench-RAG benchmark is a specific framework that addresses the retrieval-augmented generation (RAG) challenges in legal AI applications. Distinct from the broader scope of LegalBench, which appraises generative capacities, LegalBench-RAG centers on the proficiency of LLMs in accessing pertinent legal documentation and assimilating it into their responses. This benchmark is of particular importance because the ability to produce accurate and legally valid answers depends on the ability to retrieve precise legal references from large databases. The development of LegalBench-RAG was driven by the recognition of a critical gap in the evaluation of RAG systems, which are witnessing a growing use in AI-powered legal applications.

This specific benchmark scores RAGs on their ability to extract concise, relevant legal text snippets, rather than retrieving entire documents. This is a vital importance in minimizing processing costs and avoiding the risk of hallucination. By prioritizing accuracy, LegalBench-RAG helps improve the accuracy of AI-generated legal advice and supports the responsible use of LLMs in the legal field.

2.1.3 LexGLUE (Chalkidis et al., 2022)

LexGLUE (Legal General Language Understanding Evaluation) is another comprehensive benchmark designed to evaluate language models on various legal natural language processing (NLP) tasks. It aims to address the challenges inherent in processing legal texts, which often contain specialized terminology, complex syntax, and nuanced legal reasoning. LexGLUE provides a standardized framework for assessing the performance of models on a range of legal tasks, including legal judgment prediction, case law entailment, and statute law entailment. The benchmark incorporates authentic legal documents from various jurisdictions, enabling the evaluation of models on tasks such as identifying relevant legal precedents, determining entailment between legal cases, and predicting legal outcomes. LexGLUE's coverage of both civil and criminal law ensures a comprehensive evaluation of model performance in the legal domain.

LexGLUE's primary focus is to assess models based on their capacity to comprehend and process legal language with accuracy and consistency. The evaluation of models by LexGLUE is not only based on their accuracy, but also on their ability to generalize across different legal domains, highlighting strengths and weaknesses in handling complex legal texts. Utilizing standardized evaluation metrics, LexGLUE facilitates the comparison of models' performance, including LLMs such as GPT-4, BERT-based models, and specialized legal NLP systems. It has become one of the vital tools in advancing the development of legal AI models.

2.1.4 Reasoning-Focused Legal Retrieval Benchmark (Zheng et al.,2025)

The paper introduces two novel datasets, Bar Exam QA and Housing Statute QA, which are employed to evaluate retrieval-augmented generation (RAG) systems in the legal domain. The authors argue that current legal information retrieval (IR) benchmarks often rely on high levels of lexical overlap between queries and documents and lack realistic, reasoning-intensive tasks that reflect actual legal research. To address this gap, the proposed datasets include questions paired with manually annotated gold passages and answers, thereby simulating the tasks encountered by legal professionals. The Bar Exam QA dataset encompasses multiple-choice questions derived from U.S. bar exams, alongside the corresponding legal passages that have been meticulously identified through a thorough legal research process. Housing Statute QA offers Yes/No questions about eviction law across U.S. jurisdictions, annotated with relevant statutory references. The benchmark is designed to evaluate the performance of retrieval and downstream question answering (QA) by focusing on realistic legal tasks that require reasoning rather than simple keyword matching. It introduces structured legal query expansion techniques, which have been shown to significantly improve retrieval accuracy in comparison with traditional methods such as BM25 and dense retrievers. The efficacy of these enhancements is particularly pronounced due to the limited lexical overlap between legal queries and relevant documents. However, the study also finds that better retrieval does not always lead to substantial QA improvements, particularly for less capable LLMs. In summary, this benchmark presents a reasoning-focused evaluation framework utilizing carefully annotated legal datasets and highlights the need for retrieval systems that understand legal logic, not just surface-level text matching. An observation echoed in similar benchmarks like CLERC, which also emphasizes retrieval-augmented legal reasoning (Hou et al., 2024).

2.1.5 Merger Agreement Understanding Dataset (Wang et al., 2024)

The Merger Agreement Understanding Dataset (MAUD) is a large-scale, expert-annotated legal natural language processing (NLP) dataset designed to advance reading comprehension of complex merger agreements. The MAUD is based on the American Bar Association's 2021 Public Target Deal Points Study. It includes over 39,000 examples and 47,000 annotations. These were created through more than 10,000 hours of work by trained law students and attorneys. The dataset covers seven legal categories, such as Material Adverse Effect and Conditions to Closing, and provides coverage for 92 different question types. MAUD is designed to support both single-label and multi-label classification tasks, and it incorporates a technique known as rare-answer augmentation, the purpose of which is to address class imbalance.

Critically, MAUD offers baseline evaluations using widely recognized models such as RoBERTa, LegalBERT, and DeBERTa, reporting metrics like AUPR, Micro-F1, and Macro-F1 across tasks. These benchmarks establish a rigorous, reproducible framework for assessing the ability of LLMs to parse, classify, and reason about real-world contractual language. Beyond its value as a dataset, MAUD thus serves as a high-quality benchmark, laying an important foundation for evaluating and improving legal LLMs on substantive document understanding tasks in the domain of mergers and acquisitions.

2.1.6 Allens AI Australian Law Benchmark

The Allens AI Australian Law Benchmark is an evaluative framework developed by Allens, in collaboration with Linklaters LLP, to assess the proficiency of generative AI tools in providing legal advice under Australian law. Conducted in February 2024, the benchmark evaluated five prominent LLMs: GPT-4, Gemini 1, Claude 2, LLaMa 2, and Perplexity. The evaluation encompassed 30 questions across ten practice areas, including contract law, intellectual property, and data privacy, which mirror tasks typically handled by mid-level legal associates. Each model's responses were evaluated based on three criteria: substance (the accuracy of the answer), citations (relevance and correctness of legal references), and clarity (Table 2).

Substance	
0	The response is entirely **wrong**.
1	The response is generally **wrong** but contains some correct analysis.
2	The response is generally **accurate** but contains a number of errors.
3	The response is generally **accurate** but contains a small number of errors or fails to answer parts of the question.
4	The response is generally **accurate** and covers most issues.
5	The response is **accurate** and covers all material issues.
CITATIONS	
0	The citations are **fictional**.
1	The citations are **incorrect**.
2	The citations are generally **accurate** but there are important omissions.
3	Adequate and **accurate** citations are used.
CLARITY	
0	The response is very **difficult** to read.
1	The response is **clear** but not easy to read.
2	The response is **clear** and easy to read.

Table 2: Allen's Grading Scale for Legal Language Model Responses

GPT-4 emerged as the top performer, achieving an average score of 5.27 out of 10, followed by Perplexity with 4.27. LLaMa 2, Claude 2, and Gemini 1 attained scores of 3.74, 3.99, and 3.64, respectively (Figure 2).

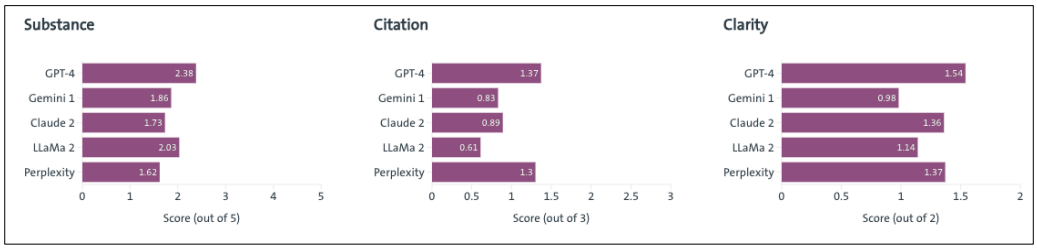

Figure 2: Model Performance Scores Across Substance, Citation, and Clarity Criteria

In spite of GPT-4's leading position, the study highlighted significant limitations across all models, particularly in tasks requiring critical reasoning and accurate legal citations. Common issues identified included incorrect, irrelevant or fictitious citations and jurisdictional errors, where models referred to non-Australian legal principles despite being explicitly instructed to do so.

The study's findings emphasize that, as of 2024, these AI models should not be relied upon for unsupervised legal advice under Australian law. While these models may facilitate the summarization of well-established legal domains, their inconsistency still requires thorough expert human review. The study also highlights the potential of specialized AI models trained on reviewed legal content and the importance of integrating AI tools.

2.1.7 Evaluation metrics and Task formats

Throughout most English-based legal AI benchmarks, they utilize a set of common evaluation metrics to objectively measure model performance. The most widely used metrics include:

- **Accuracy:** Measures the proportion of correct answers among the total, essential for classification and multiple-choice tasks.
- **F1 Score:** Combines precision and recall to provide a balanced performance measure, particularly useful for tasks with imbalanced classes.
- **Precision and Recall:** Precision measures the accuracy of positive predictions, while recall evaluates the proportion of actual positives correctly identified.
- **BLEU and ROUGE Scores:** Evaluate the quality of generated text by comparing model outputs to reference answers, frequently used in summarization and translation tasks.
- **Human Evaluation:** Often employed for subjective tasks, where experts assess the correctness and relevance of generated responses.

Additionally, the task formats commonly employed include:

- **Multiple-Choice Questions (MCQs):** Assess factual and contextual understanding.
- **Classification Tasks:** Identify categories or legal outcomes.
- **Text Summarization:** Condense legal texts into concise summaries.
- **Legal Reasoning and Case Matching:** Test the model's ability to reason through legal scenarios.
- **Extractive and Generative Question Answering (QA):** Extract specific information or generate text-based answers.
- **Document Classification and Retrieval:** Identify relevant legal documents or texts from large databases.

These metrics and formats ensure a comprehensive evaluation of LLMs' capabilities in legal contexts.

2.1.8 Chinese Legal Benchmarks

Chinese legal benchmarks for LLMs: LawBench (Wu et al., 2024), LegalAgentBench (Zhou et al., 2024), LexEval (Li et al., 2024) etc, are designed to evaluate the

performance of LLMs on complex legal tasks. In a manner very similar to English benchmarks such as LegalBench and LexGLUE, Chinese benchmarks evaluate models on legal reasoning, factual accuracy, and language comprehension. They utilize structured evaluation metrics, such as accuracy, F1 score, and human evaluation, to assess model performance across a range of legal tasks, including legal judgment prediction, document classification, and legal question answering.
However, Chinese legal benchmarks differ in focus due to the distinct legal system and language characteristics. Similar challenges are also encountered in benchmarks from other non-Western jurisdictions, where legal structures and linguistic nuances demand region-specific adaptations (Joshi et al., 2024). For instance, LexEval and LawBench, are designed for the civil law system, which is statute-based. These benchmarks frequently categorize tasks based on cognitive abilities such as memorization, logical reasoning, and legal application. Furthermore, Chinese benchmarks sometimes incorporate ethical considerations, evaluating how well LLMs can handle legal tasks without perpetuating biases.

2.2 Specialized Legal Language Models

While benchmarks are essential for evaluating the performance of legal AI systems, an equally important aspect lies in understanding the specialized legal natural language processing (NLP) repositories and models. Benchmarks depend considerably on well-constructed repositories for defining tasks, establishing evaluation standards, and simulating real-world legal scenarios. These repositories, frequently annotated by legal experts, serve as the foundation upon which benchmarks measure model accuracy, reasoning, and relevance. Simultaneously, specialized legal NLP models such as SaulLm, LawLLM, and domain-adapted versions of GPT are developed or fine-tuned using these repositories to better handle the complexities of legal language. Recent initiatives even focus on adapting LLMs to national jurisdictions (Hyperthink, n.d.). The interaction between these repositories and models guarantees that benchmarks offer not only a score but also a meaningful reflection of a model's capacity. Therefore, this section is devoted to an examination of the influential repositories and models that are closely associated with the development of benchmarks and the advancement of legal NLP.

2.2.1 SaulLM (Colombo et al., 2024)

SaulLM is a family of large-scale language models designed specifically for legal applications, making it one of the first open-source efforts tailored to understanding and generating legal text. The initial release, SaulLM-7B (Colombo et al., 2024a), was built on the Mistral 7B architecture (Smith, 2024) and trained on a large corpus of over 30 billion tokens from English-language legal materials, including those from the U.S., U.K., Canada, Europe, and limited Australian jurisdictions. In addition, SaulLM-7B-Instruct was developed as an instruction-tuned variant optimized for legal reasoning tasks through fine-tuning on synthetic dialogues and legal QA datasets.

Building on this foundation, the team later introduced SaulLM-54B and SaulLM-141B (Colombo et al., 2024b), significantly scaling up model size and training to further improve performance on complex legal tasks. These larger models aim to improve

domain adaptation and depth of reasoning in high-stakes legal contexts. Evaluation of SaulLM models has been conducted using a refined version of LegalBench, called LegalBench-Instruct, alongside the legal subset of the MMLU benchmark, targeting tasks such as issue spotting, rule recall, interpretation, and jurisprudence. In these evaluations, SaulLM models consistently outperform general-purpose LLMs such as Mistral and LLaMa, particularly in legal reasoning accuracy, while maintaining competitive perplexity scores on legal documents.

SaulLM's commitment to transparency and accessibility is underscored by the open release of its models under an MIT license, along with its evaluation tools. Overall, SaulLM's contributions are twofold: first, it provides high-performing legal-specific models; and second, it provides a scalable, replicable framework for building advanced legal LLMs based on rigorous domain adaptation.

2.2.2 LawLLM (Zheng et al., 2024)

LawLLN, an English-language LLM model developed to support the U.S. legal system by enhancing legal question answering (QA) performance with factual accuracy and domain relevance. The model underwent a two-stage fine-tuning process, initially training on synthetic data derived from legal documents and subsequently on high-quality, human-annotated data. The model is built upon GPT-like architectures and focuses on practical tasks such as answering bar exam-style questions, legal reasoning, and statute interpretation.

A distinguishing feature of LawLLM is its incorporation of realistic legal prompts, many of which mirror tasks found in legal education and practice. Its training datasets encompass a diverse array of legal materials, including statutes, case law, regulations, and bar exam content, thereby ensuring its comprehensive coverage of U.S. law. To assess its performance, LawLLM was benchmarked against multiple legal QA datasets, demonstrating superior accuracy and factual consistency when compared to other general-purpose and legal-tuned models.

A notable aspect of LawLLM is its integration of a module for factuality checking, a crucial advancement in light of the legal field's demand for precision. The architecture and data pipeline of LawLLM establishes a foundation for the development of reliable legal AI tools, positioning it as a promising basis for responsible and scalable legal natural language processing (NLP) applications in the English-speaking world.

2.2.3 Hallucination

Hallucination, where a large language model (LLM) generates information that is false, misleading, or unsupported by its source, is a critical concern in legal AI due to the high

stakes of factual and jurisdictional precision. Within the legal domain, hallucinations may take the form of fabricated statutes, case law, or erroneous legal reasoning, which can have grave ramifications if employed without human oversight.

The paper Large Legal Fictions (Zheng et al., 2024) introduces a comprehensive framework for building a profile of hallucinations in legal LLMs, including LawLLM. Study reveals that LawLLM, despite its domain-specific fine-tuning, which has led to enhanced performance on U.S. law benchmarks, still generates hallucinations in over 30% of responses across various tasks. The authors distinguish between extrinsic hallucinations (fabricated but plausible facts) and intrinsic hallucinations (inconsistent or logically incoherent reasoning). These errors frequently emerge when LawLLM seeks to generalize across legal subdomains or make unsupported interpretations of statutes, even when trained on high-quality legal dataset.

In addition, the study Legal RAG Hallucinations (Magesh et al., 2024) explores the occurrence of hallucinations in retrieval-augmented generation (RAG) pipelines utilized within legal contexts. Despite the potential of RAG to mitigate hallucinations by anchoring responses in authentic legal sources, the study revealed that leading legal research tools still exhibited hallucinations in 17–33% of queries. The prevalence of misguided citations and incongruities between legal areas undermined users' confidence in the AI-generated results.

Collectively, these studies underscore that hallucination in legal LLMs is not merely a technical nuisance but rather a systemic vulnerability. Even specialized models, such as LawLLM or advanced RAG pipelines, are vulnerable to producing unreliable outputs. To address these challenges necessitates not only the improvement of datasets and model tuning (Hu et al., 2025), but also the incorporation of legal expert oversight. Thus, to ensure that AI tools remain accurate, grounded, and ethically deployable in legal practice.

3 Summary of the State of the Art

The literature highlights the growing need for legal specialization of LLMs and jurisdiction-specific benchmarking to ensure reliable legal AI performance. Existing benchmarks like LegalBench (Guha et al., 2023), LexGLUE (Chalkidis et al., 2022), and MAUD (Wang et al., 2024) target general English legal tasks but lack Australian legal specificity. Although the Allens AI Benchmark offers early insights, its limited scope underscores a substantial gap. No comprehensive evaluation exists for Australian law contexts. Furthermore, hallucination issues, as revealed in studies like Large Legal Fictions (Dahl et al., 2024) and Legal RAG Hallucinations (Magesh et al., 2024), show the vulnerability of current models. Together, these findings reinforce the need for an Australian-focused benchmark to evaluate and guide responsible legal AI development.

Recent research increasingly emphasizes the need for LLMs to be specialised for the legal domain and evaluated through legal-specific benchmarks. Benchmarks such as LegalBench, LexGLUE, and MAUD have provided important advances in assessing legal reasoning, judgment prediction, and contract analysis using general English-language legal documents. However, these benchmarks predominantly reflect U.S. and European legal systems, leaving a critical gap in addressing country-specific legal reasoning frameworks. Similarly, emerging Chinese benchmarks such as LawBench (Wu et al., 2024), LegalAgentBench (Zhou et al., 2024), and LexEval (Li et al., 2024) have demonstrated the importance of tailoring evaluations to local legal law traditions and practices.

Despite global progress, Australia currently lacks a robust, publicly available benchmark for evaluating LLMs within its legal context. The Allens AI Australian Law Benchmark (Allens, 2024) constitutes a preliminary effort but remains limited to only 30 questions spanning a few areas of law. Its narrow scope and absence of retrieval or deep reasoning tasks restrict its utility as a comprehensive evaluation tool. As such, there is a fundamental need for a broader, more domain-focused Australian legal benchmark that reflects both federal and state law distinctions and aligns with real-world legal practices.

Compounding the lack of Australian-specific benchmarks is the broader challenge of hallucination and inconsistency in current legal LLMs. Studies such as Large Legal Fictions and Enhancing the precision and interpretability of retrieval-augmented generation (Hindi et al., 2025), have demonstrated that even advanced models like LawLLM exhibit a significant rate of factual inaccuracies, particularly when reasoning across complex legal queries. These vulnerabilities point to systemic issues that benchmarking alone can reveal but not immediately rectify. In the Australian legal context, where jurisdictional precision is critical, such hallucinations could lead to substantial professional risks.

Taken together, the literature review highlights an urgent need for a specialised, Australia-focused legal LLM benchmark that not only evaluates retrieval and reasoning performance but also systematically quantifies model reliability and hallucination tendencies. Addressing this gap forms the core objective of this thesis.

4 Research Project Plan

4.1 Introduction: Identifying the Research Gap

Large Language Models (LLMs) such as GPT-4, Claude, and Mistral have rapidly expanded into professional domains with high stakes, including the field of law. These systems are being adopted more frequently for tasks ranging from legal document summarization and statute interpretation to contract review and legal research support. However, their implementation in legal contexts necessitates meticulous evaluation tailored to specific jurisdictions, given the paramount importance of factual accuracy and jurisdictional precision in legal work.

Recent literature reveals the emergence of various legal LLM benchmarks, including LegalBench, LexGLUE, MAUD, and LegalBench-RAG, which offer comprehensive evaluation frameworks for English-language legal tasks. Despite this global progress, a significant gap remains: there is currently no standardised benchmark for evaluating LLMs within the Australian legal context. Existing benchmarks are predominantly U.S.-centric or jurisdiction-neutral, failing to account for Australia's distinctive federal-state legal structure, statutory frameworks, and judicial writing conventions.

This research project aims to address this critical gap by proposing the development of a domain-specific benchmark tailored to Australian law, with an initial focus on the field of contract law. The proposed benchmark will facilitate the evaluation of both general-purpose and legal-specific LLMs in tasks that emulate real-world Australian legal reasoning, document analysis, and statutory interpretation.

4.2 Scope and Strategic Focus

In order to ensure both feasibility and relevance, the benchmark will be delineated around three strategic dimensions: the legal domain, jurisdictional layering, and practical applicability.

4.2.1 Legal Domain Focus: Contract Law

The foundational domain for this benchmark is contract law. This decision is grounded in three justifications. Firstly, according to Progressive Legal (Aldridge, n.d.) and New South Wales Supreme Court tasks (New South Lawyers, 2024), contract disputes are among the key categories of civil litigation in Australia, reflecting their significance within the broader legal system. Secondly, contract law is extensively documented in AustLII and comprises a comprehensive array of statutory law (e.g., the Australian Consumer Law) and judicial interpretation. Thirdly, contract principles are foundational to commercial and civil law practice, making this benchmark broadly applicable to legal education, practice, and technology development.

4.2.2 Jurisdictional LayeringAustralia's federal legal system necessitates

Benchmarking that reflects the distinctions between Commonwealth law and State/Territory law. This project will account for these differences by tagging data with

jurisdictional metadata (e.g., "NSW Supreme Court," "Federal Court of Australia"). The tasks will be designed to test whether LLMs can accurately determine and apply the appropriate legal source, an essential skill for any AI model intended for legal assistance in the Australian context.

4.2.3 Practical Applicability and Realism

Rather than evaluating abstract legal reasoning, this benchmark aims to simulate tasks that legal professionals, academics, or students commonly perform. These tasks include the identification of enforceable clauses, the summarization of judicial reasoning, and the extraction or validation of case citations. The prompts and tasks will be modelled on real-world court language to ensure contextual authenticity.

4.3 Data Source and Corpus Construction

4.3.1 AustLII as Primary Corpus

The Australasian Legal Information Institute will serve as the core data source for this project. AustLII provides open-access legal materials encompassing all levels of Australian law, including court judgments, tribunal decisions, and statutory texts. AustLII is a legally reusable and academically reliable resource, offering robust metadata and structured legal text suitable for natural language processing.

4.3.2 Document and Case Selection Criteria

The benchmark corpus will be constructed from AustLII using the following criteria:

- Relevance to contract law, filtered by keywords and legal categories.
- The corpus will include both statutory excerpts (e.g., ACL, Contracts Review Act) and judgments across court hierarchies.
- Judgments will be selected based on their inclusion of substantial legal reasoning, moderate length, and citation density.
- Exclude documents with excessive procedural content or unrelated areas of law.

To automate the processes of case scraping, preprocessing, and content segmentation for benchmarking purposes, the project will be developed in consultation with an academic collaborator from the Faculty of Law, ensuring that data selection and preparation align with legal research standards and ethical considerations. To make sure consultation shapes the rules and scope, while automation executes those rules efficiently.

4.4 Benchmark Design and Task Framework

4.4.1 Multi-Task Structure

The benchmark will employ a multi-task evaluation design, integrating classification, summarization, citation, and retrieval-augmented generation (RAG) tasks. These tasks will assess the capabilities of legal LLMs across various levels of legal cognition and text comprehension.

- Binary and Multi-class Classification: For instance, the classification of clauses that result in the creation of enforceable obligations.
- Summarization: This task involves the distillation of legal reasoning in judgments into succinct summaries.
- Citation Extraction and Validation: The process entails the identification and verification of cited legal authorities.
- Open-ended Legal Prompts: Scenario-based inquiries that necessitate multi-hop reasoning and legal justification.

It is designed to simulate how legal professionals interact with source material, and all prompts and responses are based on authentic legal language. Examples of each task type, along with representative inputs and expected outputs, are shown in the table below (Table 3).

Task Type	Example Task	Input Sample	Expected Output Sample
Binary and Multi-class Classification	Determine if the following clause creates an enforceable obligation.	"The parties may, at their discretion, proceed to formalize the agreement within a reasonable time."	**No** (Language is permissive, not mandatory.)
Summarization	Summarize the legal reasoning used by the court in finding a breach of contract.	A case excerpt describing how the defendant failed to deliver services by the agreed date, causing substantial loss.	"The court found a breach due to failure to meet delivery timelines stipulated as essential terms of the contract."
Citation Extraction and Validation	Identify the primary case cited to support the court's interpretation of contract formation.	Excerpt referencing multiple cases discussing offer and acceptance.	**Carlill v Carbolic Smoke Ball Co [1893] 1 QB 256**
Open-ended Legal Prompt	Advise whether a binding agreement exists based on the following negotiation communications.	Email exchanges discussing key terms but with ongoing negotiations and subject-to-contract language.	"No binding agreement exists because essential terms were not finalized and communications were expressed to be 'subject to contract.'"

Table 3: Example Task Formulations for Contract Law Benchmark

4.4.2 Retrieval-Augmented Generation (RAG) Component

A dedicated subset of the benchmark will evaluate RAG systems, testing the ability of models to retrieve and cite relevant legal texts from a document database (e.g., a mini-AustLII corpus). The tasks will be structured to assess the following:

- The relevance and precision of retrieval.
- Correct grounding of answers in retrieved content.
- Citation accuracy and legal coherence.

RAG evaluation will adopt metrics such as Recall@K, Source Precision, and Citation Validity.

4.5 Evaluation Methodology

4.5.1 Metrics

The performance of the model across a series of standard tasks will be evaluated using metrics developed within the legal NLP community.

- Accuracy: For binary and multiple-choice question (MCQ)-based classification tasks.
- F1 Score: For multi-label classification scenarios.
- ROUGE and BLEU will be used to assess the precision of the model's predictions. For the evaluation of summarization quality.
- Citation Accuracy: The validity and relevance of referenced statutes or cases.
- Recall@K is employed to assess the accuracy of RAG retrieval.

4.5.2 Hallucination Tracking

A significant research contribution of this project is the systematic analysis of LLM hallucination in legal outputs. Utilizing established frameworks from Large Legal Fictions (Zheng et al., 2024) and Legal RAG Hallucinations (Magesh et al., 2024), responses will be annotated for:

- Extrinsic Hallucinations: Fabricated but plausible-sounding legal facts or authorities.
- Intrinsic Hallucinations: Internally inconsistent legal logic or contradictory citations.

The development of hallucination detection methods and scoring rubrics is mostly based on manual review and feedback from legal experts, sometimes with minimal automation.

4.6 Data Validation, Ethics, and Quality Assurance

4.6.1 Benchmark Quality Assurance

To ensure the reliability and validity of the benchmark, a structured quality assurance (QA) process will be implemented. Instead of generating answers through a model, the benchmark will be designed around legal materials where the correct answers are already known or can be directly derived. For instance, cases sourced from AustLII will be used where the outcome, key citations, or legal reasoning are clearly identifiable. These elements will be converted into structured tasks, such as predicting the case outcome based on a case description or identifying the relevant citation from a legal passage. Task formulation will thus inherently embed gold-standard answers. Validation of task design and question–answer alignment will be performed in consultation with legal academics or postgraduate law students, using a structured annotation and verification protocol (Dominguez-Olmedo et al., 2025). In instances where multiple interpretations are possible, explanatory notes will accompany the benchmark entry to document accepted reasoning paths and mitigate future ambiguity.

4.6.2 Ethical and Privacy Considerations

This project does not involve human participants or private data. All texts have been obtained from AustLII, a public and legally open-access repository. However, in the interest of safeguarding confidentiality, sensitive information, including names in family law or discrimination cases, will be anonymized to ensure the highest standards of privacy. Procedures for data scraping, processing, and management will be thoroughly documented to ensure transparency. The project is also subject to responsible data governance practices.

4.7 Long-Term Impact and Open Research Contributions

The objective of this project is to establish a lasting contribution to the evaluation of AI for the context of Law in Australia. By creating the first benchmark specifically tailored to evaluating LLM performance in the domain of Australian law, this work provides a critical starting point for assessing legal AI reliability in Australia. The benchmark will be supported by a technical report detailing its design and baseline assessments, as well as a systematic hallucination audit to assess and address factual and legal inaccuracies in model outputs. To ensure broad adoption and continuous improvement, the benchmark will be openly released on platforms such as Hugging Face or GitHub, accompanied by comprehensive documentation and a contributor framework. This will not only advance academic understanding, but also support the development of reliable legal AI tools, enhance legal education, and contribute to the responsible use of AI in the Australian legal context.

4.8 Timeline Gantt chart

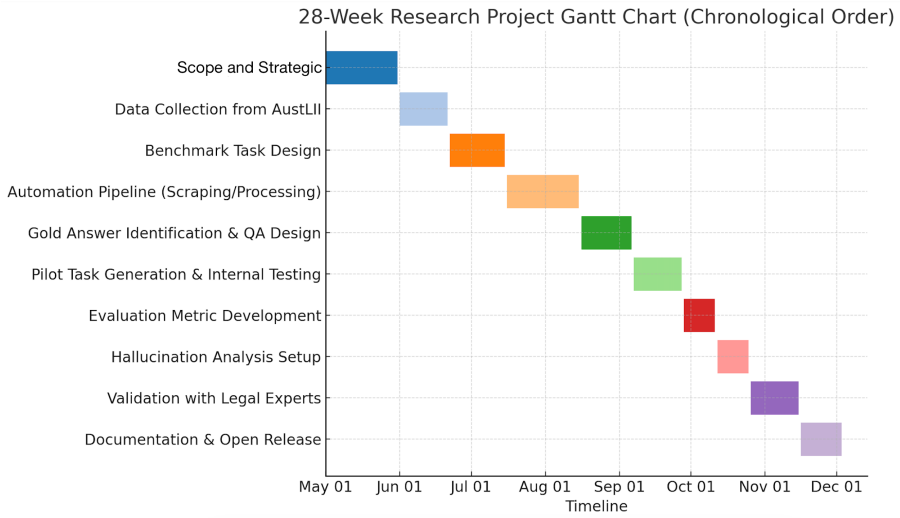

Figure 3: Preliminary Research Schedule

This 28-week timeline outlines the initial research timeline for the thesis project. It outlines a phased plan covering data collection, benchmark design, and evaluation stages. While this serves as a foundational guide, tasks and durations may be revised as the project progresses and new requirements emerge.

5 Conclusion

This research project presents a strategically grounded plan for developing a high-quality, open-access benchmark for evaluating LLMs in the context of Australian law, with an initial focus on contract law. The project is positioned to address the prevailing gap in jurisdiction-specific legal benchmarks, thereby offering a direct response to the limitations identified in extant literature. Existing frameworks, which are often generic and lack adaptation to the Australian legal environment, are a notable weakness. The proposed benchmark will feature a multi-task structure, including classification, summarization, citation analysis, and retrieval-augmented generation (RAG). These tasks will be designed by presenting cases for which the outcomes or relevant citations are already known from AustLII, and converting them into a series of questions. As generative AI continues to accelerate legal workflows (Thomson Reuters, 2024). The goal of the research is to enable more reliable, responsible, and context-aware legal AI in Australia, positioning the benchmark as a critical tool to evaluate and deploy LLMs with legal precision and integrity.

Part 2: The Research Paper

Abstract

The integration of Large Language Models (LLMs) into the legal domain presents a significant opportunity to enhance accessibility, efficiency, and analytical precision in legal practice. However, existing benchmarks such as LegalBench, LexGLUE, and MAUD remain primarily U.S. and EU centric, limiting their relevance to Australian jurisprudence. This thesis is a step towards filling this gap by proposing a benchmark for evaluating LLM performance under the Australian legal context, focusing on contract law as a foundational domain. Using authoritative materials from AustLII, comprising seven High Court judicial precedents and three key statutes,we generate 300 structured questions across multiple formats (QA, MCQ, True/False) with answers (manually checked for correctness), resulting in 1,800 experimental evaluations across LLM-only and RAG (retrieval-augmented generation) setups. Models including GPT-4o-mini and GPT-5-mini are evaluated using accuracy, F1, and LLM-as-a-Judge metrics. The results reveal that RAG configurations substantially improve factual precision, with statutory texts yielding the highest reliability and case law exhibiting variable reasoning quality. The benchmark establishes a framework for evaluation of AI for Australian jurisdiction and supports progress towards responsible adoption of LLMs for Australian legal context.

1 Introduction

The Australian legal system faces mounting challenges arising from growing caseloads, procedural complexity, and constrained resources. The line graph (Figure 1) shows the total number of pending civil and criminal cases in the Supreme Court of Victoria over nearly five years, from Q3 2021 to Q2 2025. Reported sustained increases in pending cases, raising concerns about delays in justice delivery and administrative efficiency. At the same time, the high cost of legal services continues to limit access to justice, particularly for individuals and small businesses who cannot afford professional representation (Olling, 2025). These systemic pressures have prompted interest in technological interventions capable of alleviating administrative burdens and improving decision support processes within the legal sector.

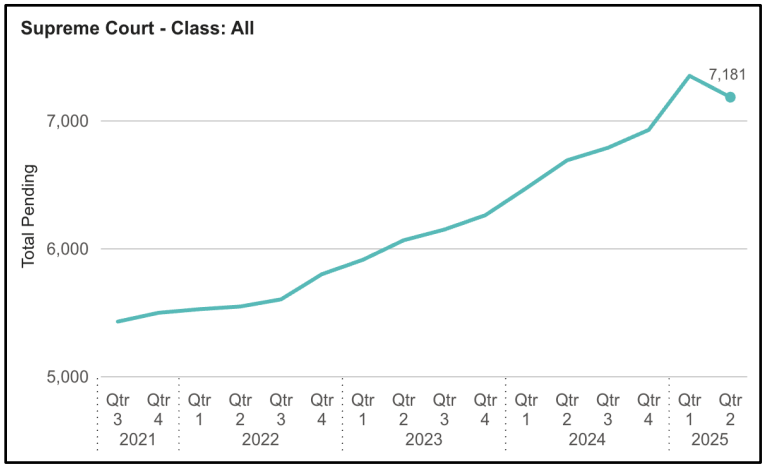

Figure 1: Victoria Supreme Court pending (source: Court Services Victoria, 2025)

In parallel, the emergence of Generative Artificial Intelligence (GenAI), particularly LLMs, has transformed the landscape of automated reasoning, document analysis, and legal information retrieval. Studies demonstrate that domain-adapted LLMs, such as LegalBERT (Chalkidis et al., 2020) and LawLLM (Zheng et al., 2024), can perform structured legal tasks such as case summarization, clause classification, and statute interpretation with notable accuracy. Despite these advances, the integration of LLMs into real legal practice requires rigorous, jurisdiction-specific evaluation frameworks. Without such evaluation, models risk generating misleading or non-jurisdictional content, problems well-documented in hallucination studies like Large Legal Fictions (Dahl et al., 2024) and Legal RAG Hallucinations (Magesh et al., 2024).

To date, international efforts to measure legal AI performance have created high-quality benchmarks including LegalBench (Guha et al., 2023), LexGLUE (Chalkidis et al., 2022), and MAUD (Wang et al., 2023). These frameworks assess models on tasks such as judgment prediction, statute entailment, and contract understanding. Yet, they remain focused on the United States or European Union, with limited relevance to Australia's unique legal structure, which integrates both common law reasoning and layered federal-state statutory interpretation. The only preliminary effort within this jurisdiction, the Allens AI Australian Law Benchmark (Allen, 2024), assessed GPT-4 and other LLMs across 30 questions spanning several practice areas. While informative, its limited scope and absence of retrieval-based tasks highlight the need for a more comprehensive, empirically grounded framework.

Addressing this gap, we develop a domain-specific benchmark for Australian contract law, designed to systematically evaluate LLM performance under realistic legal conditions. The benchmark leverages publicly accessible materials from the Australasian Legal Information Institute (AustLII), encompassing seven High Court precedents and three key statutes selected based on citation value and doctrinal coverage. Each document is semantically segmented and used to generate structured question sets across three formats, Question-Answering (QA), Multiple-Choice (MCQ), and True/False (TF), to assess both factual retrieval and interpretive reasoning. The questions and the correctness of the answers are manually checked to ensure quality.

2 Related work

Research in legal natural language processing (NLP) has expanded rapidly, yet most benchmarks remain centered on the United States and European Union. This section reviews prior scholarship across five relevant areas, legal NLP benchmarks, Australian resources, RAG for legal tasks, automated evaluation using "LLM-as-a-judge" and question generation. We then position the present work as the first Australian, contract-law-specific question-answer benchmark combining High Court cases, statutes, and evaluation of LLM-based systems to answer legal questions relevant to those cases.

2.1 Global Legal NLP Benchmarks

Seminal corpora such as LegalBench (Guha et al., 2023), LexGLUE (Chalkidis et al., 2022), MAUD (Wang et al., 2023), CaseHOLD (Zheng et al., 2021), and CUAD (Hendrycks et al., 2021) established standardized tasks including judgment prediction, legal entailment, statute interpretation, and contract clause classification. These

resources fueled the development of domain-adapted models like LegalBERT (Chalkidis et al., 2020) and LawLLM (Zheng et al., 2024).

However, their corpora overwhelmingly reflects U.S. federal or EU civil-law materials, citation norms, and drafting conventions. Australian law, rooted in common-law reasoning but mediated by overlapping federal and state statutes, differs structurally and linguistically. Hence, direct transfer of these benchmarks to Australian contexts is limited. Moreover, few existing efforts integrate both statutes and apex-court decisions, or compare closed-book and retrieval-grounded settings within one framework. This gap motivates the creation of a jurisdiction-specific benchmark for Australian contract law.

2.2 Australian Resources and Prior Benchmarks

Australia's open-access legal infrastructure, particularly AustLII and its citation engine LawCite, provides an authoritative foundation for data collection and replication. Yet systematic LLM evaluation within this ecosystem is scarce. The Allens AI Australian Law Benchmark (Allen, 2024) offered a brief 30 question evaluation across multiple legal areas, useful as an early probe but too limited to be a reliable evaluation framework . In addition to the Allens AI Australian Law Benchmark, Han, Burgess, and Shareghi introduced an Australian legal citation benchmark that focuses on citation prediction rather than Question-Answering dataset. Their work serves a different analytical purpose from Allens' evaluation and this thesis's contract-law-specific benchmark (Han, Burgess, & Shareghi, 2025).

Community resources such as the Open Australian Legal QA dataset (Butler, 2023) supplied design cues for question structure but lacked a unified evaluation pipeline. Recently, the emergence of domain-specific embeddings, for instance auslaw-embed-v1.0 trained on Australian judgments, has suggested that legal retrieval quality could improve through localized representation learning. Nevertheless, no study has directly compared such embeddings against general models in a controlled Australian benchmark. This thesis addresses that gap by combining AustLII-sourced High Court cases and statutes with side-by-side embedding evaluation, producing 1,800 scored tasks across question types and retrieval modes.

2.3 Retrieval-Augmented Generation

RAG frameworks retrieve relevant passages before generation, reducing hallucination and improving citation accuracy (Lewis et al., 2021). This approach was selected because prior studies consistently show it excels in legal contexts requiring precise source attribution and factual grounding (Zheng et al., 2025). Legal implementations typically employ dense-vector stores (e.g., FAISS) and hybrid retrievers to surface authoritative reasoning or statutory text (Panchal et al., 2025). Despite success in U.S. and EU corpora, studies note persistent issues: semantic drift, near-duplicate case confusion, and context truncation from coarse chunking (Locke & Zuccon, 2022). Mitigation strategies, document-ID filtering, re-ranking, and citation-aware prompts, have rarely been tested on Australian materials. Our benchmark extends this research by empirically comparing RAG performance on target-precedents/statutes retrieval (one source document) versus multi-precedents/statutes corpora (ten documents) within contract law.

2.4 Automated Evaluation and "LLM-as-a-Judge"

Recent work explores using large models to grade open-ended legal answers, offering scalability but raising concerns about consistency and bias (Thakur et al., 2025). Hybrid schemes combining automated scoring with selective human validation have shown strong reliability (Gu et al., 2025). Metrics such as Accuracy, Precision, Recall, and F1 remain standard for objective formats, while rubric-guided LLM judgments capture qualitative aspects like reasoning soundness and completeness. The current study follows this hybrid approach, integrating GPT-4 as a rubric-driven evaluator ("LLM-as-a-Judge") and applying human spot-checks for jurisdictional correctness.

2.5 Positioning of the Present Study

Prior work offers valuable insights but does not answer a core question for Australian practitioners: How accurately can LLMs interpret and reason with real Australian legal texts? This thesis contributes the first comprehensive, contract-law-specific benchmark that (i) unites statutes and judgments from AustLII, (ii) evaluates closed-book and RAG settings under two embedding paradigms, and (iii) reports per-document and per-question-type performance across 1,800 evaluations. Beyond empirical results, it delivers a reproducible framework to extend benchmarking into other domains such as torts, equity, administrative, and constitutional law, thereby advancing both academic research and practical AI assessment within the Australian legal system.

3 Research Methodology

3.1 Research Design

This research adopts an experimental benchmark design aimed at evaluating the capabilities of LLMs in interpreting and reasoning about Australian contract law. The benchmark is tailored to assess model performance across selected high-citation judicial precedents and key statutory instruments, focusing on legal content that is both foundational and practically significant for Australian legal professionals.

The overarching goal is to offer a qualitative and quantitative assessment of LLMs in legal contexts, helping legal experts understand current model limitations and potential. To ensure that the evaluation is both domain-specific and rigorous, the research narrows its scope to contract law, a doctrinally rich area with well-defined principles and extensive judicial interpretation.

The legal documents include seven High Court of Australia cases and three core statutes sourced from AustLII, a reliable and continually updated legal database. This ensures that the dataset is both authoritative and publicly accessible. Each document is processed and semantically chunked into smaller units. These chunks are then used as inputs for automated question generation across three formats:

1. Open-ended QA,
2. Multiple-choice (MCQ)
3. True/False (T/F).

The question generation process leverages the OpenAI API, employing gpt-4o-mini LLM with tailored prompts and programmatic control to ensure diversity and relevance

of questions. To preserve data quality and prevent hallucinations, all generated questions and answers undergo manual verification, checking for legal accuracy, relevance, and alignment with the source materials.
This research design ensures that the benchmark closely simulates the types of interpretive and reasoning tasks typically performed by human legal experts, providing a realistic and reproducible evaluation framework.

3.2 Data Collection

Our benchmarking focuses on Australian contract law, specifically using High Court of Australia (HCA) judicial precedents and key statutory instruments to construct a domain-specific benchmark for evaluating LLMs and RAG. To ensure the benchmark reflects the depth and diversity of legal reasoning within this contract domain, a multi-stage data collection process was employed.

3.2.1 Source Selection

The primary source of legal materials was AustLII1, a publicly accessible and widely trusted legal database that hosts full-text legislation and court decisions across Australian jurisdictions. AustLII's integration with LawCite2 was critical in identifying highly cited cases with significant precedential value. To optimize the search process, additional sources were used. The High Court of Australia's official judgments portal[3], which allowed keyword-specific searches and filtering by judgment date. Lexis+ Australia[4], contains the full text of Australian cases, journal articles, and online commentaries. Includes CaseBase case citator, Halsbury's Laws of Australia legal encyclopaedia, All England Law Reports, New Zealand Law Reports.

3.2.2 Case Selection Criteria

To build a high-quality and pedagogically valuable benchmark, seven judicial precedents were chosen based on the High citation value and precedential authority (Table 1), as determined through AustLII, LawCite, and citation analytics.

LawCite Citation Count	LawCite Citation Count	LawCite Level
Paciocco v Australia and New Zealand Banking Group Limited[2016] HCA 28	132	5/5
Crown Melbourne Limited v Cosmopolitan Hotel (Vic) Pty Ltd [2016] HCA 26	203	5/5
Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30	239	5/5
ZG Operations Australia Pty Ltd v Jamsek [2022] HCA 2	260	5/5
Laundy Hotels (Quarry) Pty Limited v Dyco Hotels Pty Limited [2023] HCA 6	39	5/5
Elisha v Vision Australia Limited [2024] HCA 50	19	3/5
CFMMEU v Personnel Contracting Pty Ltd [2022] HCA 1	395	5/5

Table 1. Summary of seven Selected High Court Precedents and Citation Metrics

[1] AustLII. (2025). *Australasian Legal Information Institute*. https://www.austlii.edu.au/
[2] LawCite. (2025). https://www.austlii.edu.au/LawCite/doc/overview.html
[3] High Court of Australia. (2025). https://www.hcourt.gov.au/
[4] Lexis+ Australia. (2025). https://plus.lexis.com/apac/lpauhome

Building on citation-based selection, the cases were further categorized according to their contract law subtopics, ensuring broad coverage of doctrinal diversity (Table 2).

To ensure recent relevance, priority is given to cases from 2000 onwards where legal reasoning aligns with contemporary practice and LLM benchmarking. The final list for High Court cases was curated through cross-referencing search results from AustLII, the High Courts of Australia (HCA) website, and citation-based tools (LawCite). Manual vetting ensured legal clarity, doctrinal depth, and suitability for semantic chunking and prompt engineering. In addition to case law, three key statutes were selected to reflect the legislative foundations of Australian contract law: Independent Contractors Act, Contracts Review Act, and Insurance Contracts Act. These were chosen for their doctrinal diversity, relevance to modern contractual disputes, and complementary coverage of federal and state legislation. Together, they reflect key legal dimensions such as fairness, employment status, and sector-specific regulation, ensuring the benchmark captures both foundational principles and applied complexities in Australian contract law. The final corpus for benchmarking comprises seven High Court precedents and three key statutory instruments (Table 3), reflecting both doctrinal diversity and foundational legislative coverage within Australian contract law.

High Court Case Name	**Subtopics**
Paciocco v Australia and New Zealand Banking Group Limited [2016] HCA 28	*Penalties & consumer contract fairness*
Crown Melbourne Limited v Cosmopolitan Hotel (Vic) Pty Ltd [2016] HCA 26	*Collateral contracts*
Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30	*Contract scope of the penalty doctrine*
ZG Operations Australia Pty Ltd v Jamsek [2022] HCA 2	*Employment Contract Characterisation*
Laundy Hotels (Quarry) Pty Limited v Dyco Hotels Pty Limited [2023] HCA 6	*Property sale dispute contract*
Elisha v Vision Australia Limited [2024] HCA 50	*Breach of Contract*
CFMMEU v Personnel Contracting Pty Ltd [2022] HCA 1[5]	*Employment contract law*

Table 2. Distribution of High Court Precedents Across Contract Law Themes

Type	**Legal Source (Case / Statute Name)**
High Court precedents	Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30 Paciocco v Australia and New Zealand Banking Group Limited [2016] HCA 28 Crown Melbourne Limited v Cosmopolitan Hotel (Vic) Pty Ltd [2016] HCA 26 Laundy Hotels (Quarry) Pty Limited v Dyco Hotels Pty Limited [2023] HCA 6 CFMMEU v Personnel Contracting Pty Ltd [2022] HCA 1[5] ZG Operations Australia Pty Ltd v Jamsek [2022] HCA 2 Elisha v Vision Australia Limited [2024] HCA 50
Key statutes	Independent Contractors Act 2006 Contracts Review Act 1980 No 16 Insurance Contracts Act 1984

Table 3. Final Corpus of Legal Sources Used for Benchmarking

[5] CFMMEU refers to *Construction, Forestry, Maritime, Mining and Energy Union*

3.3 Question Generation Approach

To evaluate LLM capabilities on Australian legal content, we implemented an automated question-generation pipeline using custom Python scripts for multiple-choice (MCQ), true/false (TF), and open-ended (QA) formats.

1. MCQs follow a four option format (A–D), with either single or multiple correct answers supported by legal reasoning.

2. QA questions require specific short answers grounded in statute or precedent.

3. TF questions demand binary responses validated through logical inference from the source material.

These scripts extracted text from selected statutes and judgments and applied semantic chunking, a process that divides long legal documents into smaller, meaning-preserving segments typically aligned with logical boundaries such as paragraphs, sections, or argument structures. Subsequently, we used GPT-4 via prompt engineering (Appendix B1) to generate questions aligned with legal reasoning tasks.

To ensure quality and reduce hallucinations, all outputs were manually reviewed and validated. Additionally, we referred to the Open Australian Legal QA dataset6 from HuggingFace to guide our design and calibrate question difficulty and structure against a community-developed legal QA benchmark.

While automatic legal question generation has been explored for contracts, legislation, and legal education (Deroy et al., 2025), quality control remains challenging. Weak prompts often yield vague or hallucinated content. Scholars advocate manual review and replacement loops to ensure structural consistency (Fu et al., 2024). Building on these practices, our pipeline produces semantically chunked inputs (~1,024 tokens with overlap), generates QA, MCQ, and T/F items via the OpenAI API, and manually filters low quality questions (Appendix A1 to A3). These were then replaced by regenerating new questions. before regeneration from later document segments.

The final dataset comprises 300 questions in total, with equal representation across the three distinct formats. These questions were generated from a curated corpus of Australian contract law, including 7 High Court judicial precedents and 3 key legislative instruments. The resulting 300-question dataset provides a scalable and legally aligned benchmark for evaluating LLMs in the Australian jurisdiction, specifically focused on the domain of contract law.

[6] Butler, U. (2023). *Open Australian Legal QA*. Hugging Face. Retrieved from https://huggingface.co/datasets/isaacus/open-australian-legal-qa

3.4 Evaluation Metrics

3.4.1 Accuracy, F1, Precision, Recall, LLM-as-a-judge

To evaluate LLM performance across the benchmark, we applied standard classification metrics: Accuracy, Precision, Recall, and F1-score, to quantitatively assess correctness, particularly in MCQ and TF formats where ground truth answers are clearly defined. For QA responses, which often involve free-form generation and interpretive nuance. We supplemented the evaluation with the LLM-as-a-Judge approach, employing GPT-4 with tailored prompts to assess the alignment between the answer and expected legal rationale (Appendix B2). Each response was then scored based on completeness, relevance, and factual consistency with the source.

3.4.2 Human Review and Legal Validity

In addition to automatic metrics, we incorporated manual human review to validate both the LLM answers and their underlying reasoning. Legal researchers performed quality checks on randomly sampled questions from each format, focusing on whether the model's response aligned with Australian legal standards and avoided hallucinated or irrelevant reasoning. This step was essential to ensure that evaluations reflect jurisdiction-specific legal accuracy and not merely syntactic similarity or general correctness.

3.5 Tools and Frameworks (OpenAI API, Python, Excel, LLM)

The development of this benchmark relied on an integrated set of tools and frameworks across both question generation and evaluation stages. For generating questions, Python was used in conjunction with the OpenAI API to produce structured legal prompts from court decisions and statutes. Semantic chunking techniques were applied to PDF documents using a customized semchunk module to ensure each prompt was aligned with logical sections of text. Token management was handled using tiktoken, allowing prompts to stay within LLM input limits, while openpyxl supported the structured export of questions to Excel for review.

The benchmarking phase, which involved generating answers and assessing legal reasoning quality, was built on a LangChain-based Retrieval-Augmented Generation (RAG) pipeline. The ingestion stage utilized document loaders that preserved footnotes and metadata, embedded the content with OpenAI's embedding models, and stored them in a FAISS vector database for efficient retrieval. At the querying stage, LangChain's RetrievalQA was combined with a standardized prompt template to generate responses grounded in retrieved legal context.

These tools together enabled efficient generation, retrieval, and evaluation of legal questions while supporting manual verification workflows throughout the benchmark lifecycle.

4 Experiment Design and Implementation

4.1 System Overview

The overall evaluation system was designed to simulate legal reasoning tasks under multiple conditions, using both closed-book (LLN-only) and open-book (RAG) approaches. The full pipeline spans from document ingestion and preprocessing to model querying, logging, and evaluation.

The architecture supports both LLM-only benchmarking, where the model receives only the question (or question + gold chunk), and RAG-style document-grounded evaluation, where the model must retrieve relevant context from a larger document store before generating a response. This dual-track design allows for detailed comparison between context-aware and context-unaware generation modes.

The core system relies on OpenAI models (especially gpt-4o-mini) for both question answering and evaluation (llmAsAJudge). Data is stored and retrieved using FAISS vector indexing, and query chains are managed via LangChain's RetrievalQA. All components are implemented in Python for flexibility and modularity.

4.2 LLM Configurations and Evaluation Modes

The evaluation framework in this study distinguishes between two primary modes of model usage: LLM-only and RAG (Retrieval-Augmented Generation). These modes were designed to explore both closed-book and open-book legal reasoning tasks using models from OpenAI's GPT family.

LLM-only Mode: The model is prompted with only the question, without any context or retrieval. This mode assesses the inherent legal knowledge of the model, simulating a closed-book examination.

RAG Mode: The model is supplied with additional retrieved context chunks (from one or more legal PDFs) using dense vector retrieval. This allows the model to ground its answers in real source documents, enabling legal fact-checking and document-driven reasoning.

4.2.1 Evaluated Models

GPT-4o-mini is a cost-optimized variant of GPT-4o, served as the primary engine for generating benchmark responses across all evaluation modes and question types. It was selected for its efficiency and balanced capability: the model provides rapid inference speeds suitable for large-scale experimentation while maintaining strong reasoning performance. Its architecture allows seamless handling of both generative tasks, such as open-ended question answering and true/false justification, and discriminative tasks, such as multiple-choice classification. When integrated within the RAG pipeline, GPT-4o-mini demonstrated robust contextual grounding, effectively leveraging retrieved document chunks to produce legally consistent and citation-aware answers. Moreover, the model exhibited reliable generalization across the 300 evaluation items encompassing 100 questions of each format (QA, TF, and MCQ), making it an ideal baseline for this benchmark's large-scale and multi-format experimental design.

GPT-5-mini was included in this study for performance evaluation and comparative analysis. As the newest OpenAI model available during the research period, representing the next generation of compact LLMs. Although not the primary model, it was evaluated to provide a forward-looking reference for future legal AI benchmarking. The model exhibited slightly better accuracy in Multiple-Choice and True/False questions than GPT-4o-mini, demonstrating incremental gains in factual reasoning. However, it occasionally underperformed in open-ended QA tasks due to prompt sensitivity and shorter token handling, indicating that while GPT-5-mini is more efficient, its legal reasoning consistency remains dependent on contextual input quality.

Retrieval-Augmented Generation (RAG) configurations paired GPT-4o-mini with two distinct embedding models to assess retrieval precision and semantic grounding. The first used OpenAI's ***text-embedding-3-large***[7], a general-purpose embedding model and creates embeddings with up to 3072 dimensions. Evaluated under both single-case (target-document) and multi-cases (10-document) scopes. The second employed ***adlumal/auslaw-embed-v1.0***[8], a domain-specific sentence-transformers model fine-tuned for Australian legal applications. It maps textual passages into a 384-dimensional dense vector space for high-precision clustering and semantic retrieval. The model was trained using High Court of Australia (HCA) case law from the Open Australian Legal Corpus curated by Umar Butler, excluding OCR-derived materials to preserve textual fidelity. Approximately 129,000 context–question pairs were used for training, with 14,000 for validation.

4.2.2 Benchmarking Modes

This study evaluates legal question-answering performance under two complementary benchmarking modes: LLM-only (closed-book) and Retrieval-Augmented Generation (RAG, document-grounded). These two paradigms were designed to assess both the intrinsic reasoning ability of large language models (LLMs) and their capacity to apply grounded knowledge from Australian legal sources. Together, they provide a balanced view of how models perform when reasoning independently versus when supplied with relevant contextual evidence.

In the **LLM-only (closed-book)** configuration, models were evaluated purely on the strength of their pretrained knowledge without any access to external texts or retrieval mechanisms. Each question was presented in a zero-shot setting, meaning that no additional examples, document snippets, or contextual hints were provided. This setup simulates a reasoning environment where the model must rely solely on the information encoded during training. Two models were tested under this configuration, ***GPT-4o-mini*** and ***GPT-5-mini***. The former was chosen for its computational efficiency and balanced generative capability, while the latter represents OpenAI's newest architecture, offering improved factual consistency and reasoning depth. Collectively, these models established the baseline for assessing general legal comprehension without grounding, providing a point of comparison for the RAG-enhanced setups introduced later.

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[7] OpenAI. (2025). *text-embedding-3-large* [Embedding model]. OpenAI Platform. Retrieved from https://platform.openai.com/docs/models/text-embedding-3-large

[8] Malec, A. L. (2024). *AusLaw Embedding v1.0* [Model dataset]. Hugging Face. Retrieved from https://huggingface.co/adlumal/auslaw-embed-v1.0

The **RAG (document-grounded)** configuration, by contrast, enabled each model to draw upon a curated corpus of authoritative Australian contract law documents. This corpus consisted of ten legal PDFs, seven High Court of Australia (HCA) judgments and three key statutes, representing a comprehensive spread of doctrinal and interpretive issues. The RAG pipeline retrieves semantically relevant text chunks from this corpus using dense vector search before appending them to the model's prompt, enabling grounded reasoning supported by explicit references to real legal materials.

To evaluate retrieval and reasoning quality, the benchmark employed two embedding strategies: OpenAI's text-embedding-3-large, a high-performing general-purpose model, and adlumal/auslaw-embed-v1.0, a domain-specific model fine-tuned on Australian legal corpora. The latter demonstrated stronger contextual alignment and superior discrimination between closely related cases, such as Andrews v ANZ (2012) and Paciocco v ANZ (2016). Both embeddings were indexed using FAISS to facilitate high-precision retrieval of the top-k relevant segments.

4.3 Execution Pipeline

The benchmarking process was structured as a modular and automated pipeline, enabling efficient processing of legal documents, embedding, storage, and retrieval-based answering using different LLMs. It integrates pre-processing, embedding, and querying components into a single reproducible workflow, allowing scalability across multiple document sets and model configurations. The design supports both LLM-only and RAG modes, with traceable logs for reproducibility and evaluation, ensuring transparent benchmarking model performance across diverse legal datasets.

4.3.1 Preprocessing

A preprocessing module implemented using PyMuPDF[9] to extract legal text while preserving the integrity of doctrinal and citation-based references. The script systematically processes each PDF page, reconstructing lines to avoid fragmented text and isolating small-font regions near the bottom of pages as potential footnotes. These notes are parsed, mapped, and re-inserted into the body text as inline markers to maintain logical connections between reasoning and cited authorities.

4.3.2 Embedding and Storage

Each document chunk was converted into high-dimensional vector representations using either OpenAI's text-embedding-3-large model or the domain-specific adlumal/auslaw-embed-v1.0, a Hugging Face embedding trained on Australian legal corpora. The latter was optimized to capture nuanced statutory language and citation relationships, offering stronger contextual grounding for jurisdiction-specific reasoning. All embeddings were normalized and stored locally within a FAISS index, enabling rapid semantic similarity search and scalable retrieval across multiple documents. The pipeline incorporated token-aware batching and de-duplication to ensure efficient memory use and consistent indexing.

[9] PyMuPDF. A Python library for programmatic PDF text extraction, analysis, and document manipulation. https://pymupdf.readthedocs.io/en/latest/

4.3.3 Querying and Logging

The querying phase was handled byLangChain's RetrievalQA framework. Upon receiving a question, the pipeline retrieved the top-k relevant chunks using FAISS, and forwarded the prompt to the OpenAI LLM (e.g., gpt-4o-mini, gpt-5-mini). Responses were logged in structured format alongside retrieval metadata and model scores. Separate scripts were used for closed-book zero-shot baselines and GPT-based evaluation respectively.

4.4 Evaluation Procedure

The evaluation strategy was designed to comprehensively assess the performance of both LLM-only (closed-book) and RAG (retrieval-augmented generation) modes across different question formats and embedding strategies. The process prioritized accuracy, fairness, and reproducibility, and was split into multiple stages to facilitate both answer generation and automated grading.

4.4.1 Question Bank Generation

For each of the selected High Court decisions and statutes, 30 questions were generated, 10 per type: Multiple Choice Questions (MCQ), True/False (TF), and Short Answer (QA). These questions were written to test:

- Legal comprehension (e.g., obligations, rights, definitions),
- Factual recall (e.g., dates, amounts, parties),
- Reasoning ability (e.g., application of statutory language or judicial logic).

Each question was associated with the corresponding chunk index and source page to support benchmarking consistency.

4.4.2 Answer Generation

Two main modes of answer generation were used:

- LLM-only (Zero-shot):
  Each question was directly passed to the model (e.g., gpt-4o-mini, gpt-5-mini) without any additional context. This evaluated the model's ability to answer based solely on prior training.

- RAG (Document-grounded):
  The top-k most relevant chunks (retrieved via FAISS index) were appended to the prompt. Answers were generated using the same LLMs with contextual support. Two embedding strategies were tested:

  - OpenAI's text-embedding-3-large
  - adlumal/auslaw-embed-v1.0 (domain-tuned)

4.4.3 Scoring and Grading

In this study, each generated answer was evaluated against a gold reference using two distinct approaches, depending on the question type. For multiple-choice (MCQ) and true/false (TF) questions, an automated metric method was used, where answers were compared by exact match to the predefined correct options. This ensured objective scoring consistency for binary and discrete-answer formats. In contrast, open-ended question-answering (QA) responses required a more interpretive approach, where GPT-4 functioned as an independent evaluator under the LLM-as-a-Judge framework. This model compared the generated response to the gold reference and assigned a score based on three main criteria: factual correctness (alignment with the verified legal answer), legal soundness (consistency with reasoning in the relevant statute or case), and completeness (the extent to which the response addressed the question fully and accurately). All scores were normalized across question types to enable fair averaging and cross-format comparison. The grading rubric illustrated in Table 4 defines the scoring scale in detail.

Question Type	Criteria	Score
True/False	Answer is correct	1
	Answer is incorrect	0
Multiple Choice (MCQ)	Full correct set selected	1
	Exactly 1 error (missed 1 correct or added 1 extra invalid)	0.5
	Two or more errors (missed 2+ correct or added 2+ extras)	0
Short Answer	perfect match	1
	mostly correct, small gaps	0.7 - 1
	partially correct	0.4 - 0.7
	minimal correctness	0.1 - 0.4
	incorrect	0 - 0.1

Table 4. Scoring Rubric for Benchmark Evaluation Across Question Types

5 Results and Analysis / Report

This section presents the experimental findings of the benchmarking study, comparing LLM-only and RAG configurations across multiple legal reasoning tasks. It highlights model accuracy, retrieval precision, and interpretive reliability, followed by visual and tabular analyses of performance across question types and legal sources. The section also examines comparative model outcomes, retrieval metrics, and confidence intervals to evaluate consistency and robustness. Finally, it identifies key patterns, discusses observed limitations, and connects these insights to broader implications for evaluating large language models in the Australian legal domain.

5.1 Overview of Results

The evaluation results show clear performance trends across both LLM-only and RAG configurations. Overall, RAG systems consistently outperformed pure LLM models across all question types, including multiple choice (MCQ), true/false (TF), and open-ended question answering (QA). This demonstrates the benefit of retrieval-grounded context in supporting legal reasoning tasks.

As shown in Table 5, the **AusLaw-embed-v1.0** model with the one-case retrieval setup achieved the highest average accuracy (0.842), followed closely by the ten-case configuration (0.826). In contrast, the baseline **LLM-only** models, *GPT-4o-mini* and *GPT-5-mini*, recorded substantially lower averages (0.658 and 0.690, respectively), reflecting their limited ability to infer precise legal facts without contextual grounding. Although these closed-book with zero-prompt models are faster to deploy and computationally efficient, their lack of access to case-specific materials restricts their interpretive accuracy in nuanced legal reasoning. The **text-embedding-3-large** model also demonstrated strong retrieval performance (up to 0.802 on average), validating its robustness in dense semantic search tasks. With **1,800 benchmarked questions**, these results confirm that AusLaw-embed-v1.0 provides the most reliable and domain-aligned representation for Australian legal contexts, establishing it as the preferred embedding model for contract law reasoning and retrieval-based legal AI systems.

Table 6 further breaks down results by High Court cases and statutes, revealing that Insurance Contracts Act 1984 and Contracts Review Act 1980 achieved the highest mean performance (0.835 and 0.802). These outcomes suggest that models handled statutory material slightly more effectively than long-form judicial reasoning, possibly due to clearer logical structures in legislation. In contrast, Laundy Hotels (Quarry) Pty Ltd v Dyco Hotels Pty Limited [2023] HCA 6 and Elisha v Vision Australia Limited [2024] produced the lowest overall accuracy (0.691 and 0.698), likely reflecting the complex contractual interpretation and multi-party context of judicial reasoning.

Collectively, these findings confirm that RAG-based methods offer measurable improvements in factual grounding and interpretive precision, particularly within the Australian legal domain. They also highlight the influence of document complexity and embedding strategy on model performance, setting the stage for deeper comparative and error analyses in subsequent sections.

Evaluation Configuration	MCQ score	TF score	QA score	Avg
LLM_ZeroPrompt_gpt-4o-mini	0.730	0.73	0.513	0.658
LLM_ZeroPrompt_gpt-5-mini	0.820	0.75	0.501	0.690
RAG_text-embedding-3-large (to 1 case)	0.905	0.87	0.632	0.802
RAG_text-embedding-3-large (to 10 case)	0.855	0.82	0.636	0.770
RAG_auslaw-embed-v1.0 (to 1 case)	0.860	0.91	0.756	0.842
RAG_auslaw-embed-v1.0 (to 10 case)	0.860	0.89	0.728	0.826

Table 5. Overall Performance of LLM and RAG Configurations Across Question Types[10]

Case name	Tex-embed (1pdfs)	Tex-embed (10pdfs)	Auslaw-embed(1pdfs)	Auslaw-embed(10pdfs)	gpt-4o-mini	gpt-5-mini	AVERAGE
Andrews v Australia and New Zealand Banking Group Limited [2012]	0.860	0.753	0.913	0.860	0.510	0.680	0.763
CFMMEU v Personnel Contracting Pty Ltd [2022]	0.800	0.783	0.900	0.863	0.657	0.740	0.791
Crown Melbourne Limited v Cosmopolitan Hotel (Vic) Pty Ltd [2016]	0.730	0.760	0.767	0.797	0.697	0.717	0.745
Elisha v Vision Australia Limited [2024]	0.840	0.687	0.757	0.750	0.623	0.533	0.698
Laundy Hotels (Quarry) Pty Limited v Dyco Hotels Pty Limited [2023]	0.633	0.590	0.793	0.787	0.700	0.643	0.691
Paciocco v Australia and New Zealand Banking Group Limited [2016]	0.840	0.717	0.893	0.813	0.807	0.730	0.800
ZG Operations Australia Pty Ltd v Jamsek [2022]	0.760	0.827	0.823	0.807	0.540	0.663	0.737
Contracts Review Act 1980 No 16	0.840	0.857	0.860	0.860	0.670	0.727	0.802
Independent Contractors Act 2006	0.843	0.863	0.850	0.860	0.620	0.683	0.787
Insurance Contracts Act 1984	0.877	0.867	0.863	0.863	0.753	0.787	0.835

Table 6. Overall Performance by High Court Cases and Statutes[11]

[10] In Table 5, green indicates the highest performance values, red is the lowest.
[11] In Table 6, dark green indicates the highest performance values, light green the second; Dark red the lowest, and light red the second lowest.

5.2 Quantitative Performance Metrics

This section extends from the outcome-oriented overview in Section 5.1, introducing the key interpretive metrics that make the benchmark explainable and analytically grounded. this section focuses on *why* those results occurred through quantifiable indicators. It quantitatively evaluates model behaviour using standardized retrieval and scoring metrics. It is divided into two complementary analyses, retrieval efficiency and system-level reliability, each offering interpretable indicators of model precision, consistency, and generalizability within the benchmark.

5.2.1 Retrieval Performance (RAG Only)

Including retrieval performance is essential because it quantifies how effectively a model retrieves legally relevant information, a critical foundation for grounded reasoning in RAG systems. Across all task formats, the domain-specific AusLaw-embed-v1.0 consistently surpassed the general-purpose text-embedding-3-large in retrieval quality (Figure 2). AusLaw achieved notably higher precision (avg 0.826 vs 0.77), indicating superior ability to distinguish semantically close yet legally distinct contexts, such as differentiating penalty clauses from liquidated damages. Despite near-perfect recall (> 0.95 in TF tasks) observed in both models, AusLaw maintained a more balanced Recall–Precision relationship, leading to higher F1@K scores (avg 0.899 vs 0.864). This stability was most evident in the QA task, where nuanced and interpretive queries benefited from its domain-tuned embeddings, ensuring substantively grounded rather than lexically similar retrieval.

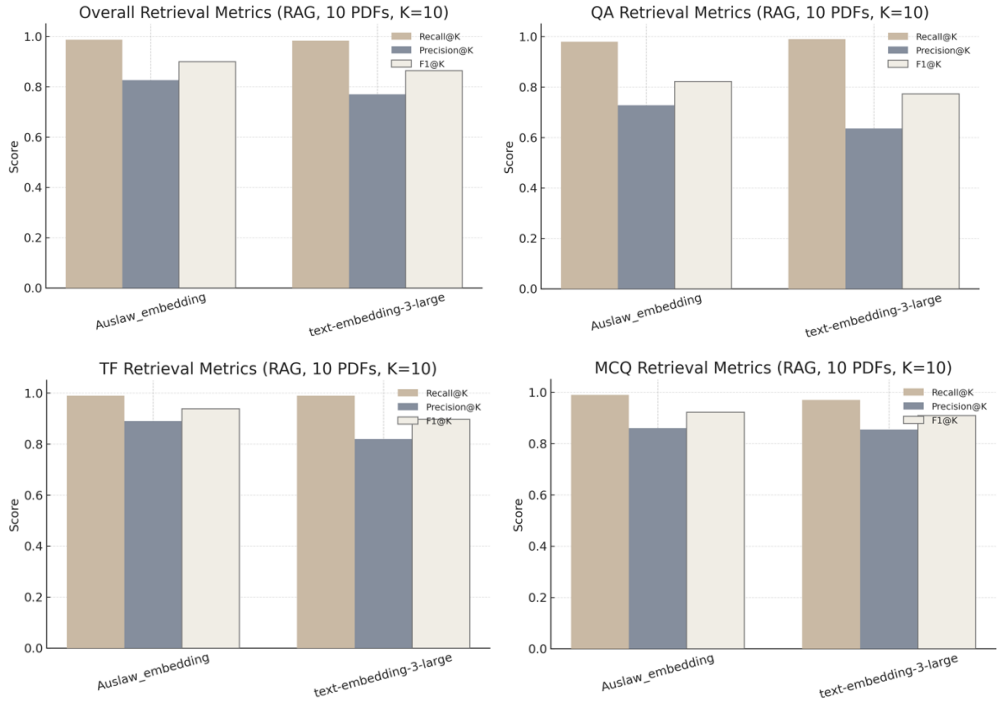

Figure 2: Retrieval Performance Across Question Types.

5.2.2 Aggregated Performance (Across Systems)

To capture cross-model consistency, macro and micro averages with **95% confidence intervals** were computed (Figure 3).

- **Macro averages** measure mean accuracy per question type, reflecting robustness across heterogeneous task structures.
- **Micro averages** aggregate all responses, emphasizing overall correctness weighted by sample size.

The narrow confidence bands in both charts (< ± 0.03) indicate stable behaviour across 1,800 benchmark instances. AusLaw-embed again leads (macro 0.83, micro 0.82), followed by text-embedding-3-large (≈ 0.79). LLM-only baselines (GPT-4o-mini and GPT-5-mini) showed broader variance and lower means, highlighting the impact of external grounding on reliability.

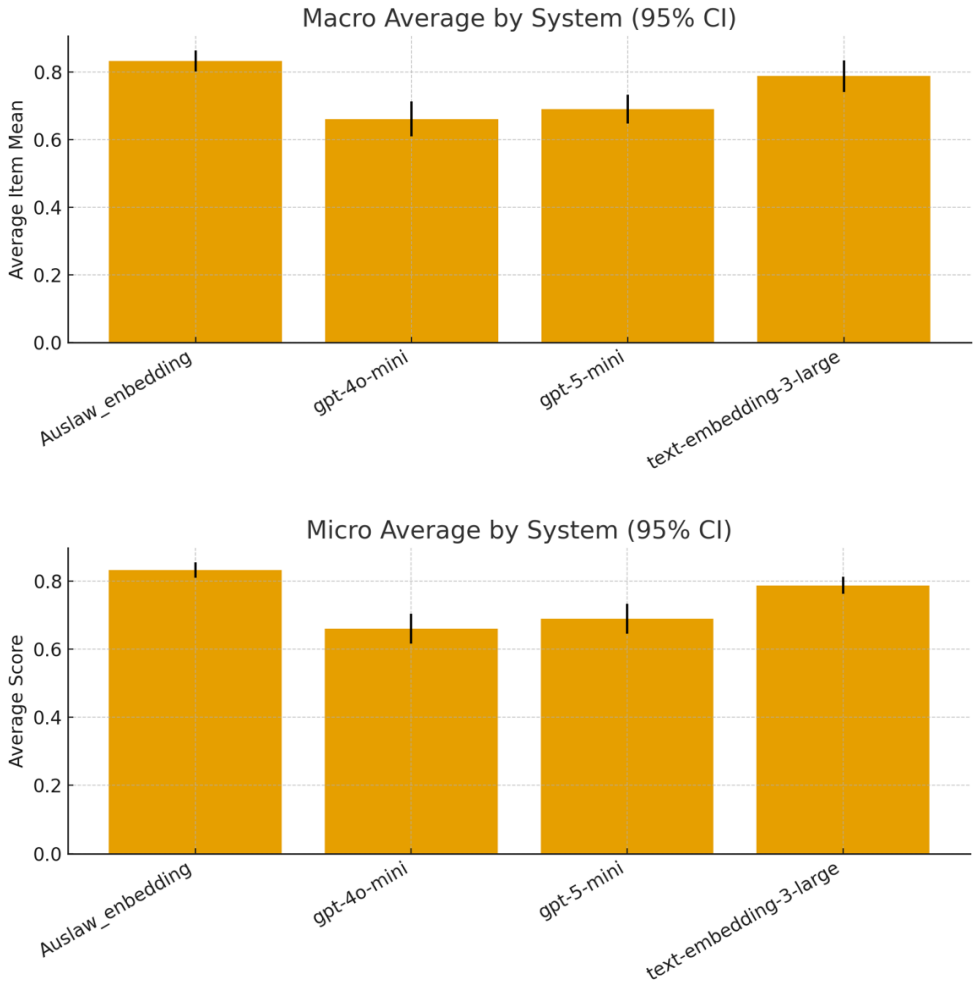

Figure 3: Model-Level Macro and Micro Averages with 95% Confidence Intervals

5.3 Comparative Analysis

This section presents a comparative performance analysis across three axes: question type (QA, MCQ, and True/False), evaluation mode (LLM-only vs RAG), and retrieval scale (single target case vs 10-case corpus). Each configuration was evaluated using 30 questions per legal document, 10 per format, drawn from High Court decisions and statutory sources.

Across all configurations, **Multiple-Choice Questions (MCQ)** produced the highest accuracy, followed by **True/False (TF)** and **Question Answering (QA)**, confirming that models perform better when constrained by structured options. QA tasks, requiring open-ended reasoning and textual interpretation, remained the most difficult for both LLM-only and RAG modes due to higher sensitivity to prompt phrasing and limited grounding context.

Comparing **LLM-only** and **RAG** modes, the inclusion of document retrieval consistently enhanced performance across all question types (Figure 4). GPT-4o-mini, for instance, achieved an overall score of 0.658 in closed-book mode, which increased to 0.802 when paired with text-embedding-3-large in the single-case RAG setup. However, expanding to a 10-case corpus slightly reduced precision (average = 0.770), suggesting marginal interference from less relevant contexts. Nonetheless, RAG again demonstrated stronger performance when handling case-grounded reasoning tasks, though the inclusion of more documents introduced additional contextual noise, indirectly affecting overall accuracy. When contrasting embedding models, the **AusLaw-embed-v1.0** model demonstrated superior and stable performance across both single and multi-document retrieval settings (average ≥ 0.82), surpassing the general-purpose text-embedding-3-large. This stability highlights the benefits of domain-adapted embeddings trained on Australian legal corpora, which better preserve semantic precision in nuanced legal contexts.

We also illustrate these differences across individual High Court cases and statutes (Figure 5). Interestingly, no strong correlation was found between document size (e.g., Elisha v Vision Australia Limited 42 KB vs Contracts Review Act 1980 127 KB) and model performance, suggesting that the determining factor is not volume but the clarity and structure of legal reasoning. Statutory texts tended to yield higher consistency due to explicit clause boundaries, whereas long judicial opinions with embedded arguments introduced interpretive ambiguity.

Overall, the findings affirm that **RAG-based pipelines, especially those using AusLaw embeddings, deliver more precise and contextually grounded legal reasoning** than closed-book LLMs. They also underscore that retrieval scale and corpus relevance critically affect downstream accuracy, a key consideration for future legal AI system design.

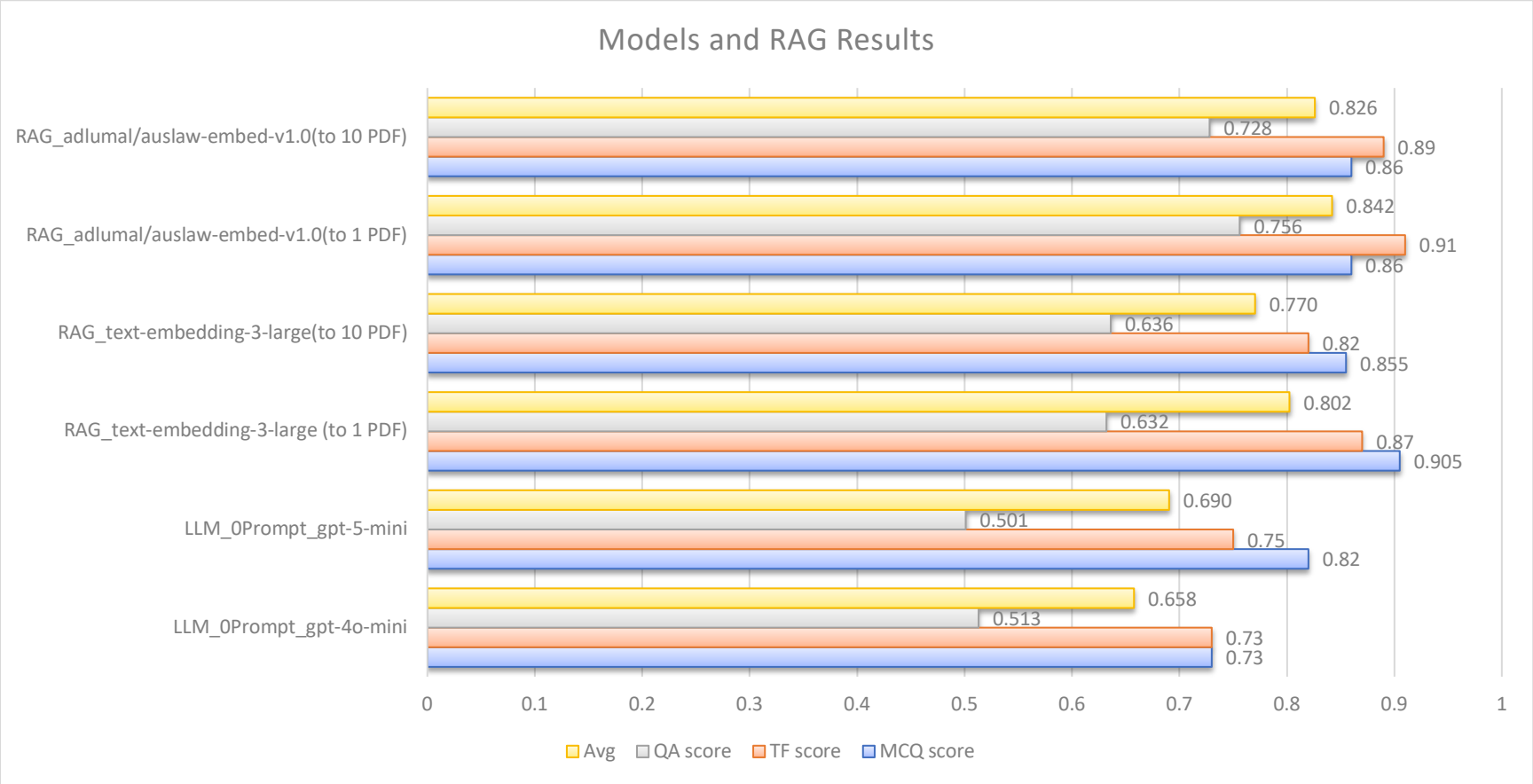

Figure 4: Comparative Performance of LLM and RAG Configurations Across Question Types

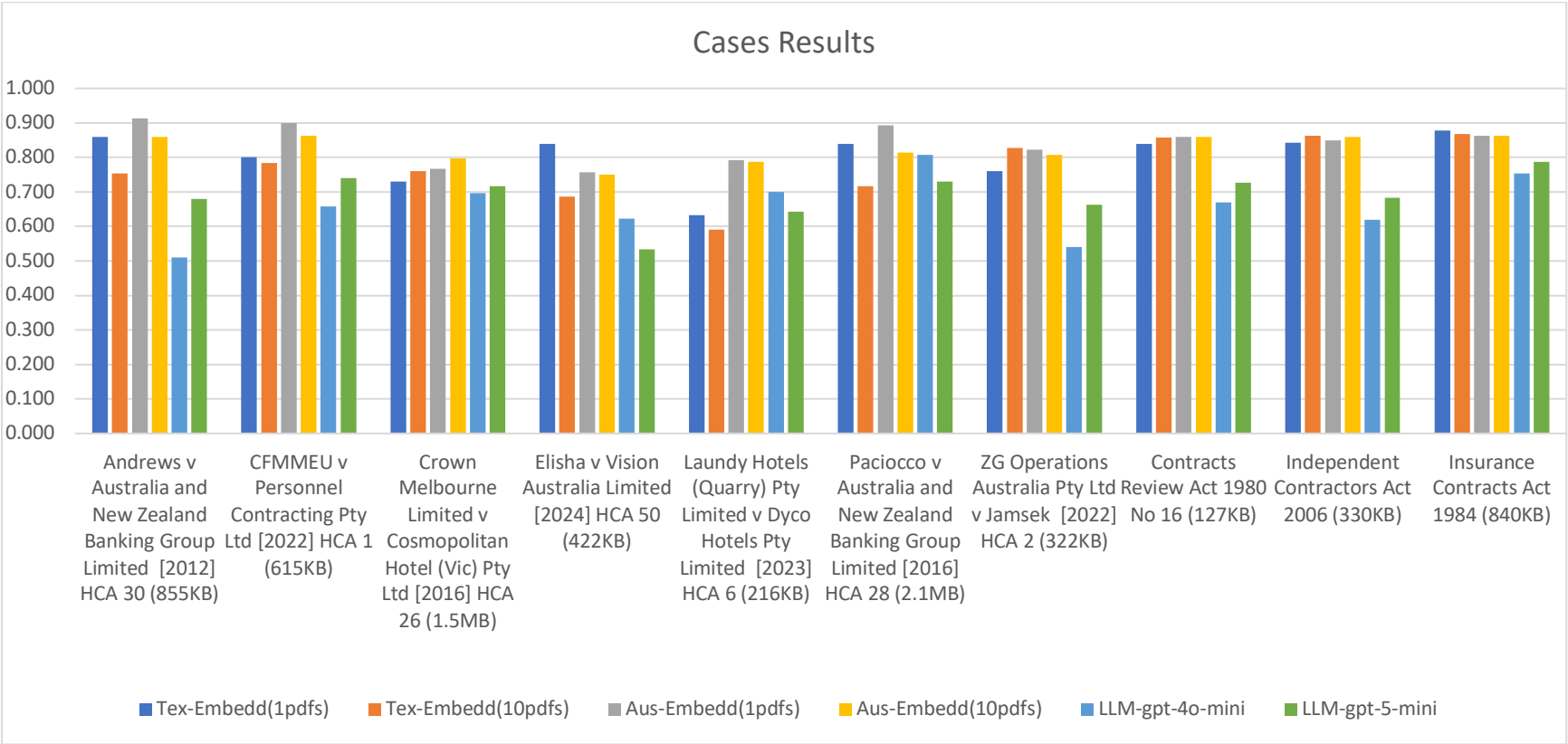

Figure 5: Comparative Results by High Court Cases and Statutory Materials

5.4 Error Analysis (e.g., hallucinations, misinterpretations)

While the overall accuracy of both LLM-only and RAG-based approaches shows promising results across question types, several types of errors were observed during the evaluation that merit closer scrutiny. These errors generally fall into three categories: citation hallucinations, misinterpretations, and false negatives due to failure to retrieve or acknowledge known content.

A notable source of error in the LLM-only runs, particularly in the QA format, stemmed from the lack of access to document-specific content, resulting in generic disclaimers or refusals to answer. For example, GPT-5-mini frequently responded with statements such as, "I don't have access to that judgment text…" or "Please paste the relevant passage…" when prompted with case-specific questions like the response of Mr. Elisha in Elisha v Vision Australia Limited [2024] HCA 50, or Ms. Hauser's perception in the same case. These outputs, while arguably safer in avoiding hallucination, still count as incorrect in benchmark scoring since the answers were indeed present within the evaluated cases.

Interestingly, RAG methods were far less likely to reject questions due to missing context. However, they introduced a different class of error, source misattribution. For instance, when asked to explain the distinction between a penalty and a pre-estimate of liquidated damages in Andrews v ANZ [2012] HCA 30, the system instead cited Paciocco v ANZ [2016] HCA 28 as the source, likely due to semantic overlap and shared entities (i.e., both cases involved ANZ and the penalty doctrine). While technically related, this undermines the goal of precise legal grounding and demonstrates the need for enhanced disambiguation in multi-document retrieval settings.

Additionally in some cases, the correct answer was available in the retrieved corpus but not selected by the model, even under the 10-PDF RAG configuration. For example, in a multiple-choice question asking, "At what ages did Mr. Jamsek and Mr. Whitby leave high school?", the RAG model responded that the context did not provide this information. In reality, the answer did exist in the case content for ZG Operations Australia Pty Ltd v Jamsek [2022] HCA 2, indicating that either the chunk was not retrieved or its salience was underestimated by the LLM.

These limitations highlight a trade-off between precision in document access (RAG) and fluid language generation with no grounding (LLM-only). In aggregate, RAG systems were less prone to refusal or irrelevant elaboration, but could still introduce subtle legal inaccuracies if retrieval retrieved semantically similar but legally distinct cases. These findings reinforce the importance of careful chunk design, improved embedding models, and fine-grained citation supervision in future benchmark iterations.

For full transparency, failed answers, including unsupported rejections, false negative, and citation mismatches are recorded in the Appendix C. This includes a row-level breakdown per legal source (e.g., HCA 1, HCA 6, Contracts Review Act 1980 No 16) to support future debugging and model refinement.

5.5 Implications for Legal AI in Australian Context

Compared to Allen's Australian Law Benchmark released in May 2024, which included only 30 total questions across various legal domains, this research represents a significantly more comprehensive and fine-grained assessment focused exclusively on the domain of Australian contract law. It not only increases the number of evaluated items by a factor of 60 but also provides detailed breakdowns across multiple task formats and model types. This allows for far more nuanced insights into LLM performance, especially under grounded (RAG) vs. unguided (LLM-only) conditions.

The broader implication of this work lies in its practical reference value for legal professionals, and researchersin Australia. As LLMs become more integrated into legal workflows, a robust, domain-specific benchmark like the one developed here enables practitioners to better understand model behavior, strengths, and failure modes within real-world legal scenarios.

By surfacing the variance in performance across different legal documents, embedding strategies, and question types, this benchmark helps demystify the capabilities and limitations of current LLMs. It also encourages informed adoption of LLM-based tools for legal research, document review, or case summarization, ultimately fostering a responsible and evidence-driven approach to AI adoption in the legal field.

This benchmark is intended as a foundation for future work. The methodology and results can support ongoing iterations, such as benchmarking newer LLMs, integrating Australian-specific embeddings, or extending the evaluation framework to other legal domains such as torts, equity, or administrative law. Ultimately, this work aims to bridge the gap between legal expertise and technical capability, ensuring that legal professionals in Australia have access to meaningful, transparent, and rigorous benchmarks when evaluating LLM tools.

6 Conclusion

6.1 Key Findings and Their Significance

This study demonstrates that Retrieval-Augmented Generation (RAG) models significantly outperform LLM-only baselines in contract law question answering, particularly in complex QA tasks. Embedding quality and context size influence performance, with Auslaw-specific embeddings yielding the highest accuracy. While LLMs show acceptable baseline performance, their outputs lack legal depth and often fail to cite relevant sources. The benchmark’s granularity, spanning multiple formats and legal documents, provides a much-needed lens into how legal AI behaves in structured legal evaluation. This elevates both technical and legal discourse in Australia by offering the most detailed contract law-focused benchmark to date.

6.2 Limitations of the Current Study

While comprehensive, this benchmark is limited by its exclusive focus on contract law, which may not reflect LLM performance across broader legal domains. The static document selection and reliance on pre-generated questions constrain variability and adaptability. Furthermore, open-ended QA grading, though consistent, remains partly subjective. Some failure cases may stem from ambiguous phrasing or subtle legal distinctions that even human readers could misinterpret. Also, the evaluation did not explore model behavior under adversarial prompts or changes in prompt structure. Finally, real-world utility remains to be validated in legal workflows beyond academic benchmarking.

6.3 Recommendations for Future Benchmarks

Future legal AI benchmarks should expand to include multiple legal domains such as tort, administrative, and constitutional law. Integrating adversarial, multi-hop, and argumentative question types can test deeper reasoning. Evaluation should incorporate diverse Australian legal sources, such as state-based rulings and tribunal decisions, to test jurisdictional robustness. Human-in-the-loop grading or expert annotation could help mitigate subjectivity in QA scoring. On the systems side, future benchmarks should systematically sweep core RAG/LLM hyperparameters: retrieval **k** (and adaptive k), chunk size/overlap and windowing, hybrid retrieval (BM25 + vectors) with re-ranking (e.g., cross-encoder) and reciprocal rank fusion, context length limits and deduplication, and prompt templates (citation-required, answer+quote, chain-of-thought off/on). Model decoding knobs, temperature, top-p, and logit bias, should be calibrated for each task type (MCQ/TF/QA). Add citation verification and source attribution constraints to reduce hallucinations; test answer formatting (JSON schemas) for auto-grading. Compare embedding models with distance metrics, and evaluate footnote/figure-aware parsing. Lastly, Dynamic retrieval experiments and multilingual evaluations would reflect real-world scenarios. To support open research and reproducibility, the benchmark code and datasets have been released publicly (Appendix D)

6.4 Ethics/Data Privacy

Research was conducted in compliance with Monash University's ethical research standards[12] and data management policies[13]. All materials used in the benchmark were publicly available through verified legal databases such as AustLII and the High Court of Australia website. No personal, confidential, or client-identifiable data were collected, stored, or processed. Model outputs were anonymized and reviewed solely for academic evaluation. All dataset handling, including text chunking and embedding storage, was performed locally to ensure compliance with data privacy and integrity principles, maintaining full transparency and reproducibility throughout the benchmarking process.

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[12] Monash Research Ethics & Integrity. https://www.monash.edu/graduate-research/support-and-resources/resources/research-ethics-and-integrity

[13] Monash Data management. https://www.monash.edu/library/help/assignments-research/graduate-research/data-management

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Part 3: Appendices

Appendix A

Appendix A.1 Evaluation of Low-Quality **MCQ** Items with Diagnostics

Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30	
Which of the following cases is referenced in the case Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30?	Ambiguous citation request.
In the case of Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30, which of the following cases was referenced?	Ambiguous citation request.
Which of the following references is mentioned in the case Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30?	Ambiguous citation request.
CFMMEU v Personnel Contracting Pty Ltd [2022] HCA 1	
In the case of Construction, Forestry, Maritime, Mining and Energy Union v Personnel Contracting Pty Ltd [2022] HCA 1, which of the following references were mentioned in the snippet?	Ambiguous citation request.
Elisha v Vision Australia Limited [2024] HCA 50	
In the case of Elisha v Vision Australia Limited [2024] HCA 50, what was one of the key issues under consideration?	Over-broad prompt ("one of the key issues") admits many correct summaries
Independent Contractors Act 2006	
According to the Independent Contractors Act 2006, which of the following is a part of the Act?	Vague scope —unbounded and not tied to any section or schedule.
Insurance Contracts Act 1984	
According to the Insurance Contracts Act 1984, which provision discusses the interpretation of regulations?	Mis-specified concept
According to the Insurance Contracts Act 1984, which of the following is not a provision included in the Act?	Over-broad and carve-out question
According to the Insurance Contracts Act 1984, to which of the following does the Act not apply?	Over-broad and carve-out question.
ZG Operations Australia Pty Ltd v Jamsek [2022] HCA 2	
Which of the following cases is referenced in the case ZG Operations Australia Pty Ltd v Jamsek [2022] HCA 2?	Ambiguous citation request.
Which case is referenced in the provided snippet from the case ZG Operations Australia Pty Ltd v Jamsek [2022] HCA 2?	Ambiguous citation request.

Appendix A.2 Evaluation of Low-Quality **QA** Items with Diagnostics

Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30	
Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30, what is the duty mentioned in the snippet?	Question lacks specificity
Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30, which other cases or legal documents were referenced?	Citation question ambiguous
CFMMEU v Personnel Contracting Pty Ltd [2022] HCA 1	
In the case of Construction, Forestry, Maritime, Mining and Energy Union v Personnel Contracting Pty Ltd [2022] HCA 1, what are the specific references made to the 2020 case?	Question is vague and does not specify which legal references
Contracts Review Act 1980 No 16	
What is the name of the Act discussed in the case of Contracts Review Act 1980?	Question too narrow
Crown Melbourne Limited v Cosmopolitan Hotel (Vic) Pty Ltd [2016] HCA 26	
In the case of Crown Melbourne Limited v Cosmopolitan Hotel (Vic) Pty Ltd [2016] HCA 26, what was the main contractual issue under consideration?	Use of "main" issue is imprecise; multiple contractual aspects could qualify
Elisha v Vision Australia Limited [2024] HCA 50	
In the case of Elisha v Vision Australia Limited [2024] HCA 50, what was the concern regarding Mr. Elisha's behavior?	Question is underspecified; lacks contextual framing
Independent Contractors Act 2006	
Under the Independent Contractors Act 2006, what provisions can regulations make?	Question lacks focus
Under the Independent Contractors Act 2006, when did the provisions of the Act, not covered elsewhere in the table, commence?	question too procedural; "when commenced" without contextual grounding
Insurance Contracts Act 1984	
What is the purpose of the Insurance Contracts Act 1984?	"Purpose" question overly general
Paciocco v Australia and New Zealand Banking Group Limited [20 6] HCA 28	
In the case of Paciocco v Australia and New Zealand Banking Group Limited [2016] HCA 28, what was the issue regarding late payment fees?	Overly broad; does not clearly define the fee or clause
In the case of Paciocco v Australia and New Zealand Banking Group Limited [2016] HCA 28, what was the nature of the provision for payment of a fee that was being examined?	Question misaligned with textual evidence
ZG Operations Australia Pty Ltd v Jamsek [2022] HCA 2	
In the case of ZG Operations Australia Pty Ltd v Jamsek [2022] HCA 2, which sections of the FW Act, SGA Act, and LSL Act were referenced?	Section-based question too technical without interpretive framing.
In the case of ZG Operations Australia Pty Ltd v Jamsek [2022] HCA 2, what were the financial arrangements between the partnerships of Mr Jamsek and Mr Whitby and ZG Operations Australia Pty Ltd?	question is factual recall without analytical depth

Appendix A.3 Evaluation of Low-Quality **True/False** Items with Diagnostics

Laundy Hotels (Quarry) Pty Limited v Dyco Hotels Pty Limited [2023] HCA 6	
In the case of Laundy Hotels (Quarry) Pty Limited v Dyco Hotels Pty Limited [2023] HCA 6, the Public Health (COVID-19 Restrictions on Gathering and Movement) Order (No 2) 2020 (NSW) and Public Health (COVID-19 Restrictions on Gathering and Movement) Order (No 3) 2020 (NSW) were referenced.	Ambiguous citation request
CFMMEU v Personnel Contracting Pty Ltd [2022] HCA 1	
In the case of Construction, Forestry, Maritime, Mining and Energy Union v Personnel Contracting Pty Ltd [2022] HCA 1, was cited in the 2020 volume of the Federal Court Reports at page 631.	Question too specific (at page 631)
Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30	
In the case of Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30, references the case Aid/Watch Inc v Federal Commissioner of Taxation (2010) 241 CLR 539 at 549 [22]; [2010] HCA 42.	Ambiguous citation request
In the case of Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30, the court referenced the case of Rolfe v Peterson (1772) 2 Bro PC 436 at 442 [1 ER 1048 at 1052].	Ambiguous citation request
In the case of Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30, the Forestry Commission (New South Wales) v Stefanetto (1976) 133 CLR 507 was referenced. True or False?	Ambiguous citation request
In the case of Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30, the Judiciary Act 1903 (Cth), s 80 was referenced. True or False?	Ambiguous citation request
ZG Operations Australia Pty Ltd v Jamsek [2022] HCA 2	
In the case of ZG Operations Australia Pty Ltd v Jamsek [2022] HCA 2, the case of Whitby v ZG Operations Australia Pty Ltd [2018] FCA 1934 was referenced.	Ambiguous citation request
In the case of ZG Operations Australia Pty Ltd v Jamsek [2022] HCA 2, the case of Whitby v ZG Operations Australia Pty Ltd [2018] FCA 1934 was referenced.	Ambiguous citation request

Appendix B

Appendix B.1 Questions Generate Prompt Engineering

The prompt template used for creating MCQ questions

```
prompt = f"""{case_context}Based on the snippet below, generate ONE multiple-choice
question with exactly 4 options.

<snippet>
{snippet}
</snippet>

# Format
You must format your response as follows:
<format>

# Question
{{The question here}}

# Options
(A) {{option A}}
(B) {{option B}}
(C) {{option C}}
(D) {{option D}}

# Correct Answer
{{A/B/C/D}}
</format>

# Instructions
- Your question must be decontextualised, standalone, and must reference the full case name
exactly as provided above.
- There should be more than two correct answers. Include all correct options and clearly
supported by the snippet.
- The incorrect answers should be plausible but not supported by the text.
- Do not use "All of the above" or "None of the above" as options.
"""
```

The prompt template used for creating Question-Answering questions

prompt = f"""# Snippet
{case_ctx}The snippet from an Australian legal document from which you must synthesise a question and an answer is below.

<snippet>
{snippet}
</snippet>

Format
Return exactly:

Question
In the case of Insurance Contracts Act 1984, what is the duty imposed on parties to a contract of insurance regarding good faith?

Answer
{{A concise answer, extracted from the snippet, that stands alone without the snippet.}}
"""

The prompt template used for creating True/False questions

prompt = f"""{case_context}Based on the snippet below, create ONE True/False style question. Then, provide a verbatim excerpt from the snippet that directly supports your answer.

<snippet>
{snippet}
</snippet>

Format
You must format your response as follows:
<format>
Statement
{{A factual or interpretive statement based on the snippet.}}

Answer
{{True or False}}
Explanation
{{A sentence or passage exactly copied from the snippet that justifies the answer.}}
</format>

Instructions
- The statement MUST be **false** given the snippet.
- The Answer MUST be exactly **False**.
- Your statement must start with the full case name exactly as provided above (including the square-bracket citation if present).
- The answer must be decontextualised and standalone.
- Use only the snippet as your source. """

Appendix B.2 LLM-As-A-Judge Prompt Engineering

The prompt template used for LLM-As-A-Judge
RUBRIC_PROMPT = """ You are a strict evaluator in Legal content. Compare the following two answers and give a score between 0 and 1. Rubric: - 1.0 = Perfect match (e.g., answer reproduces all key legal principles completely and accurately). - 0.7~1.0 = Mostly correct, small gaps (e.g., identifies correct duty but omits a detail like timing or scope). - 0.4~0.7 = Partially correct (e.g., mentions the correct Act but misstates some provisions or misses key principles). - 0.1~0.4 = Minimal correctness (e.g., vague reference such as "duty of good faith" without explanation). - 0~0.1 = Incorrect (e.g., talks about unrelated law such as property rights). Instructions: - Only return a numeric score between 0 and 1. - Do not explain, just output the score. Answer A (ground truth): {ground_truth} Answer B (model output): {model_answer} """

Appendix C Record of Model Failures and Unsupported Responses

GPT-5-mini unsupported rejection answers
I don’t have access to external databases right now, so I can’t pull up the judgment text to confirm the precise order. Would you like me to look it up now? If so I can retrieve the decision and give you the exact ruling and the relevant paragraph references.
I can’t determine that from here — I don’t have the source citation or link. Please provide the document or where the Act was accessed so I can check.
Cannot determine. I cannot verify web access dates (no browsing access to confirm the 13 May 2025 access).
I don’t have external internet access in this chat to retrieve the case text. Would you like me to look up the judgment and extract the exact terms of the leases (I can fetch and summarize them if you permit)?
I don't have that specific VCAT finding memorized. Would you like me to look it up and provide the exact VCAT conclusion (with citation)?

RAG citation mismatch

Question: In the case of Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30, what was the court's view on the distinction between a stipulation upon default for the transfer of property and a payment of money?
A) The court held that there is a clear distinction between the two.
B) The court held that such a distinction would elevate form over substance.
C) The court held that the distinction is irrelevant in the context of penalty clauses.
D) The court did not comment on the distinction between the two.

Answer: B C (C is incorrect)
The court indicated that distinguishing between a stipulation upon default for the transfer of property and a payment of money would elevate form over substance, and it also noted that the distinction is irrelevant in the context of penalty clauses.

Sources (incorrect): Paciocco v Australia and New Zealand Banking Group Limited [2016] HCA 28.pdf p.24; Paciocco v Australia and New Zealand Banking Group Limited [2016] HCA 28.pdf p.52

Question: In the case of Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30, which of the following statements were made by the Court of Appeal?
A) The modern rule against penalties is a rule of law, not equity.
B) The limits of the doctrine of penalties arise from the consequences of breach of contract.
C) The public policy of keeping commercial parties to their bargains is reflected in the doctrine of penalties.
D) The doctrine of penalties is a rule of equity, not law.

Answer: B C (Correct option A is missed)
The Court of Appeal recognized that the limits of the doctrine of penalties arise from the consequences of breach of contract and acknowledged the public policy of keeping commercial parties to their bargains.

Sources (incorrect): Paciocco v Australia and New Zealand Banking Group Limited [2016] HCA 28.pdf p.24; Paciocco v Australia and New Zealand Banking Group Limited [2016] HCA 28.pdf p.52

Question: In the case of Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30, the ANZ appealed against the primary judge's finding that the late payment fee was payable upon breach of contract and therefore was capable of characterisation as a penalty. True or False?

Answer: FALSE
The ANZ appealed against the primary judge's finding that the late payment fee constituted a penalty, not that it was payable upon breach of contract.

Sources **(incorrect)**: Paciocco v Australia and New Zealand Banking Group Limited [2016] HCA 28.pdf p.43; Paciocco v Australia and New Zealand Banking Group Limited [2016] HCA 28.pdf p.7

Question: Andrews v Australia and New Zealand Banking Group Limited [2012] HCA 30, what is the distinction between a penalty and a pre-estimate of liquidated damages?

Answer: A penalty imposes an additional detriment for non-observance of a primary contractual stipulation, while a pre-estimate of liquidated damages is intended to reflect the genuine loss suffered due to a breach of contract. This distinction is highlighted in the context of the High Court's holding that a stipulation is prima facie a penalty if it imposes an additional detriment not aligned with a genuine pre-estimate of damages.

Sources (incorrect): Paciocco v Australia and New Zealand Banking Group Limited [2016] HCA 28.pdf p.24; Paciocco v Australia and New Zealand Banking Group Limited [2016] HCA 28.pdf p.6

RAG False Negative
Question: In the case of ZG Operations Australia Pty Ltd v Jamsek [2022] HCA 2, what is known about the educational background of Mr Jamsek and Mr Whitby? **Answer** (Incorrect):There is no information provided about the educational background of Mr Jamsek and Mr Whitby in the case of ZG Operations Australia Pty Ltd v Jamsek [2022] HCA 2. The context focuses on their employment history and contractual arrangements rather than their education. **Ground-Truth Evidence**: "Mr Jamsek and Mr Whitby **left high school at 14 and 15 years of age** respectively." (Source: ZG Operations Australia Pty Ltd v Jamsek [2022] HCA 2)
Question: In the case of Elisha v Vision Australia Limited [2024] HCA 50, how did Ms Hauser perceive Mr Elisha's reaction? A) Mr Elisha showed remorse. B) Mr Elisha seemed shocked. C) Mr Elisha was indifferent. D) Mr Elisha was angry. **Answer** (Incorrect): C) Ms Hauser perceived Mr Elisha as showing no remorse and being quite comfortable with the idea that the staff member had fabricated the incident. **Ground-Truth Evidence**: "[Ms Hauser] thought [Mr Elisha] presented himself very arrogantly and showed no remorse. [Ms Hauser] thought [Mr Elisha] **seemed shocked** to be told that we would be considering the matter further and he wouldn't be going back to work." (Source: Elisha v Vision Australia Limited [2024] HCA 50)

Appendix D Repository and Benchmark Data Access

The benchmark evaluates multiple LLM configurations and retrieval strategies to assess legal reasoning accuracy within Australian contract law. The code and data developed in this project are available at the project's public GitHub repository[14].

[14] GitHub Repository: https://github.com/AlanTseng0012/Australian-Legal-LLM-Benchmark